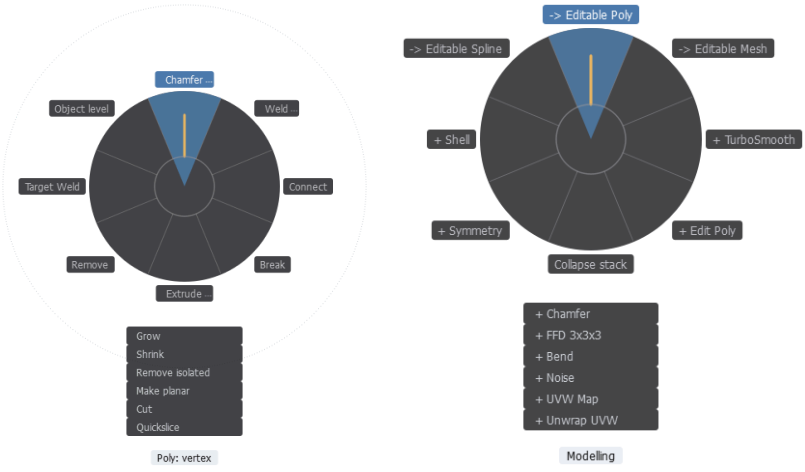
MarkingForge

QuickStart - the long version

3DS MAX 2027 · VERSION 1.5.60 · ABOUT 40 MINUTES



Every lesson, in plain words. 22 lessons: lessons 1-6 teach you to use the dials, lessons 7-22 show how to change them. Every picture is the real program. Installing is in the **Installation Guide**; every option is in the **Reference**. All of them are in **MarkingForge > Documentation**.

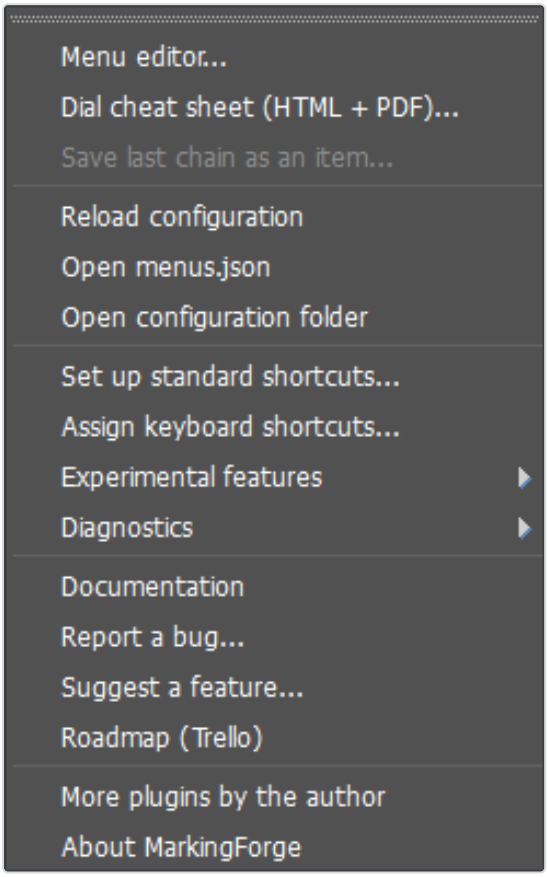
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7. How to learn it, and good habits
8. If something goes wrong
9. Support, bug reports and feature requests

Before you start

Check two things. It takes ten seconds.

1. Look at the 3ds Max menu bar. Is there a **MarkingForge** menu? If not, see the Installation Guide.
2. The dials need keys. The first time 3ds Max starts with MarkingForge, each dial gets its usual key by itself - but only if nothing else uses that key yet. To check, open **MarkingForge > Set up standard shortcuts...** It shows the keys it would add, or tells you every dial has one.



The MarkingForge menu. Everything in this guide starts here or on a key.

Your 40 minutes

Minutes	What you do
0 - 10	Lessons 1-6: how to use the dials.
10 - 15	Lesson 7: what is where in the editor. Lesson 8: your first own command on a dial.
15 - 40	Lessons 9-22: everything else you can change.

Six ideas in one minute

1. A dial is a round menu under a key

Hold Alt+1 and eight commands appear around your mouse. There are 24 dials: Alt, Ctrl+Alt and Shift+Alt, each with the numbers 1 to 8.

2. You aim, you do not read

Move towards a command and let go of the key. Your hand remembers the direction. After a week you will not need to look.

3. Hold, flick or tap

Hold the key to see the dial. Press and move at once (a *flick*) and the command runs without the dial even showing. Let go in the middle to cancel.

4. The dial follows your work

The same key shows vertex tools when you work on vertices, edge tools on edges, spline tools on a line - and material tools over the Slate editor.

5. You can change anything

In the menu editor you can put any 3ds Max command (there are over 4000), a ready-made tool, a smaller dial or a number slider in any direction. One key can even hold several dials, like pages you turn with the mouse wheel. No restart needed.

6. It is safe to try

Cancel is always one move away. Ctrl+Z undoes every command. The editor changes nothing until you press Save.

PART 1

Using the dials

Six lessons that teach your hand the dials MarkingForge comes with. Open an empty scene and try each step as you read.

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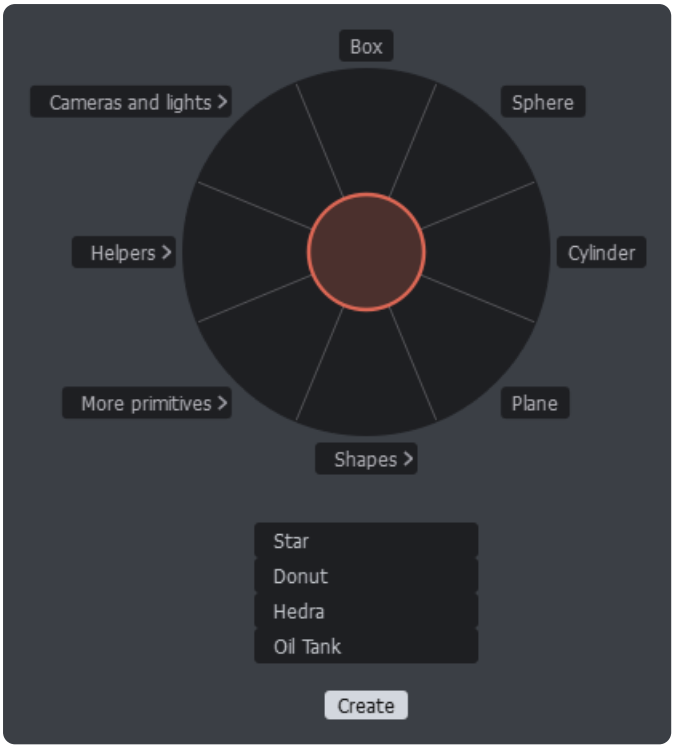
1 Your first gesture

2 min

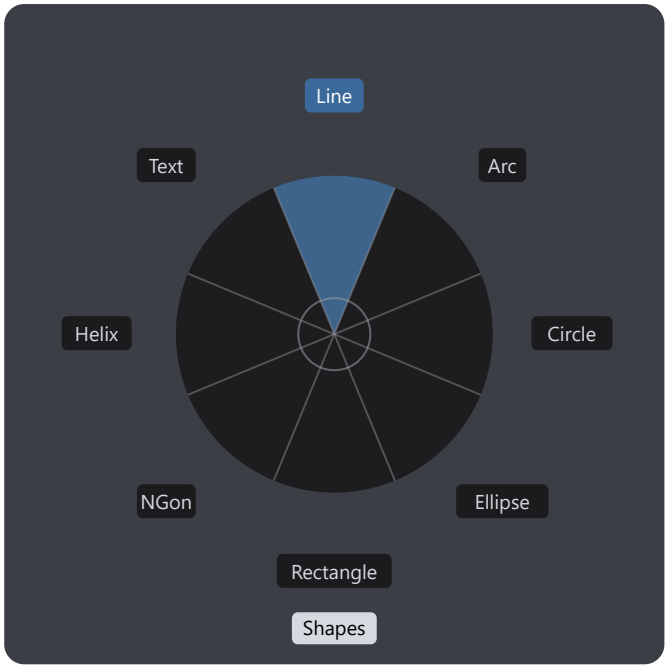
A **gesture** is simple: hold a key, move the mouse in a direction, let go.

1. Put the mouse over a viewport.
2. Hold **Alt+4**. The **Create** dial appears around the mouse.
3. Move up, towards **Box**. That part of the dial lights up.
4. Let go. You can now drag a box in the viewport, as usual.
5. Hold **Alt+4** again and move **down**, through **Shapes >**. Do not let go. The dial stays in the same place, but now it shows shapes. Move up to **Line** and let go. Draw a line; Esc finishes it.
6. Click the box to select it. Hold **Alt+1** over it, move up to **-> Editable Poly** and let go.

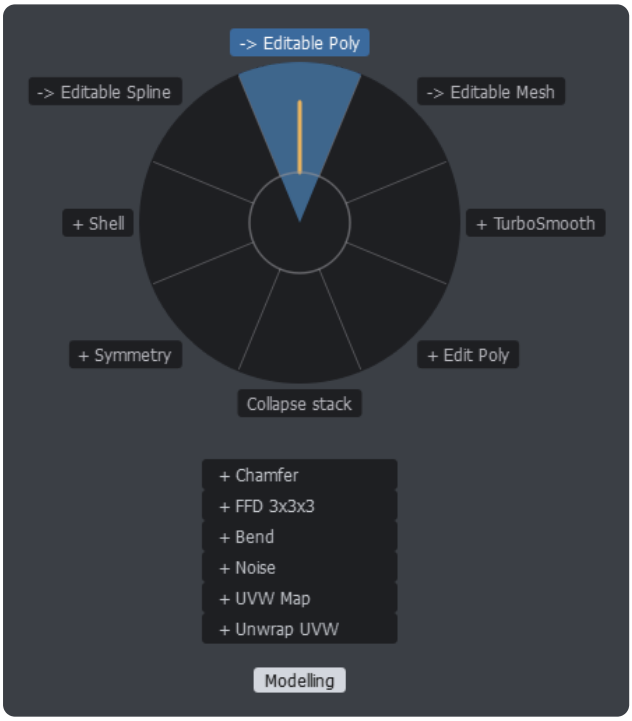
Submenu - a caption ending with **>**. It holds more commands. Move through it without letting go, and the dial shows what is inside.



Alt+4 - Create. Up and to the right: four solids. Down and to the left: four groups, each a submenu (marked >).



Inside Shapes, aimed at Line.



Alt+1, moved up: the box becomes an Editable Poly.

RESULT

The box is an Editable Poly - look at the modifier stack. Two gestures, no menu, no command panel.

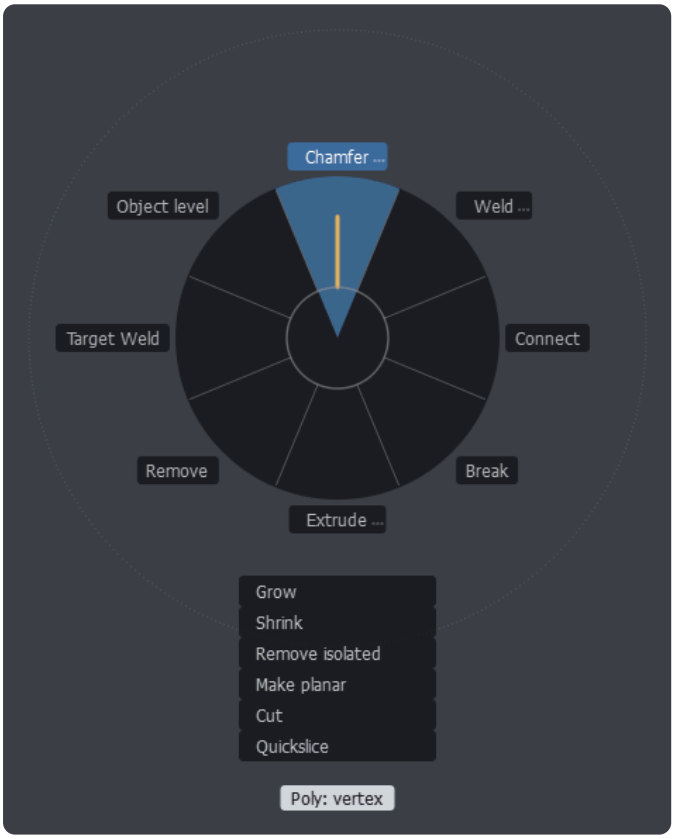
Changed your mind? Move back to the middle and let go, or press Esc. Nothing runs.

2 The dial follows you

4 min

Sub-object level - the part of the object you work on: vertices (key 1), edges (2), borders (3), polygons (4), elements (5).

1. Keep the Editable Poly selected. Press **1** - now you work on vertices.
2. Hold **Alt+1**. The same key now shows vertex tools: Chamfer, Weld, Connect, Break...
3. Move up to **Chamfer**, let go, then drag in the viewport.
4. Press **2** (edges), select an edge and hold **Alt+1**: Connect, Chamfer, Loop, Ring...
5. Press **4** (polygons), select a polygon, hold **Alt+1** and move right: **Inset**.
6. At any level, move up-left: **Object level** takes you back to the whole object.



Vertices, aimed at Chamfer.



Polygons, aimed at Inset.

Learn it once. Chamfer, Extrude, Weld... are in the same place on an Editable Poly and on an Edit Poly modifier.

RESULT

You worked on three levels of the object with one key.

3 Flick - the fast way

3 min

The dials are fastest when you do not wait for them. Press the key and move at the same moment, then let go. The command runs and the dial never shows. This is a **flick**.

TRY IT

Do each one five times - slowly first, then without looking at the dial:

Press, then move	What happens
Ctrl+Alt+7, up	Zoom so you see everything
Ctrl+Alt+7, down	Show only what is selected (do it again to go back)
Alt+3, up / down	Select all / select nothing
Ctrl+Alt+1, up / right	Move tool / scale tool
Alt+7, up	Undo one step

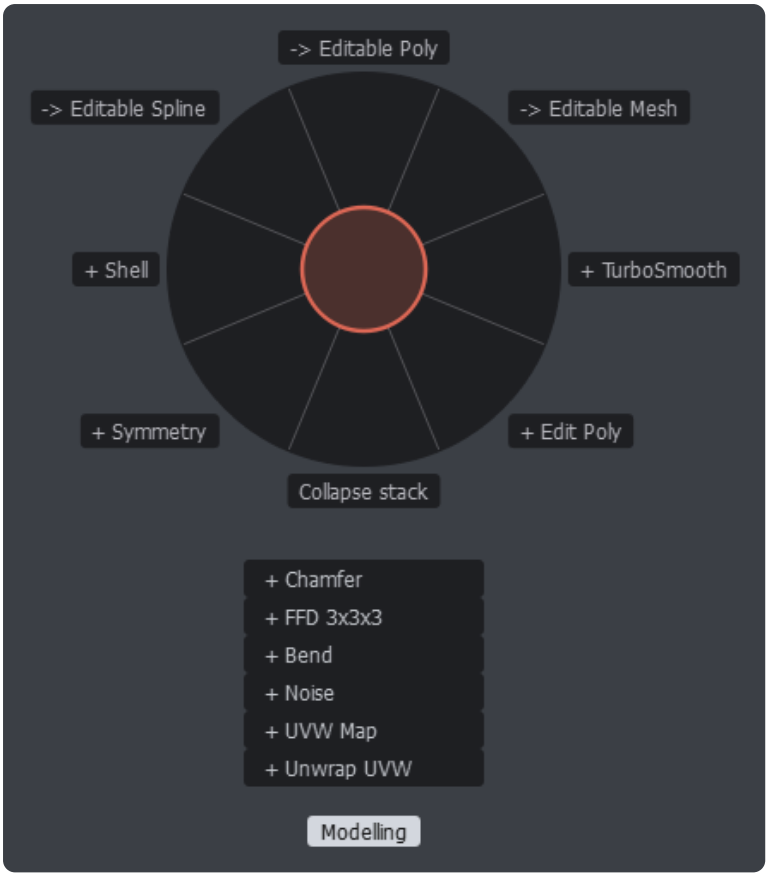
RESULT

A day later you will do these five without looking. That is the whole idea - the other dials work the same way.

4 More than eight - rows, three dots and the dotted circle

3 min

Some dials have **rows under them**. They hold commands you use, but not often enough to give them a direction. Keep holding the key, move down to a row and click it.



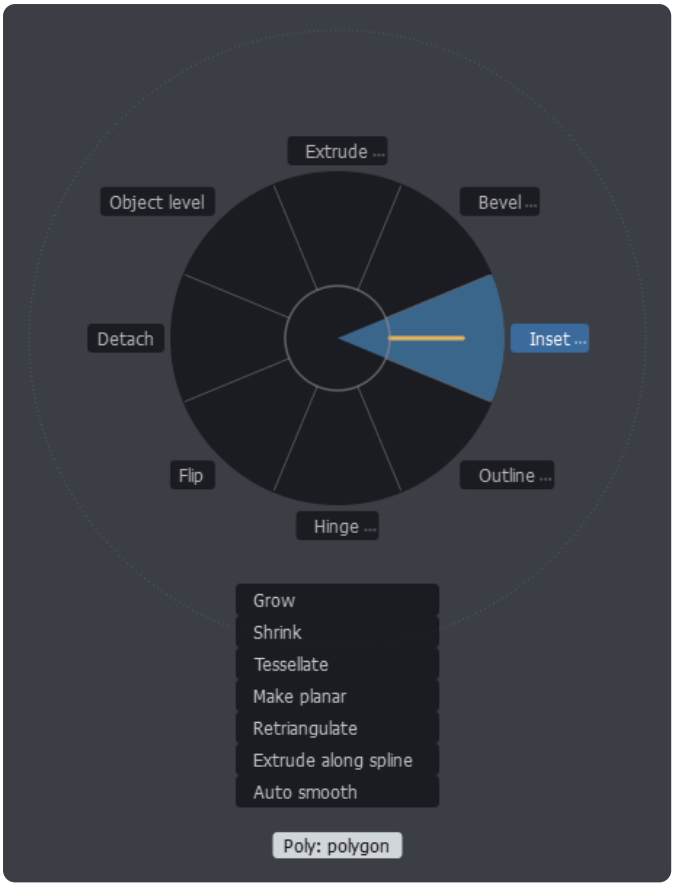
Alt+1 on an ordinary object: eight directions and six rows (Chamfer, FFD, Bend, Noise, UVW Map, Unwrap UVW).

Some captions end with **three dots** - Chamfer, Extrude, Inset... These commands have a settings window.

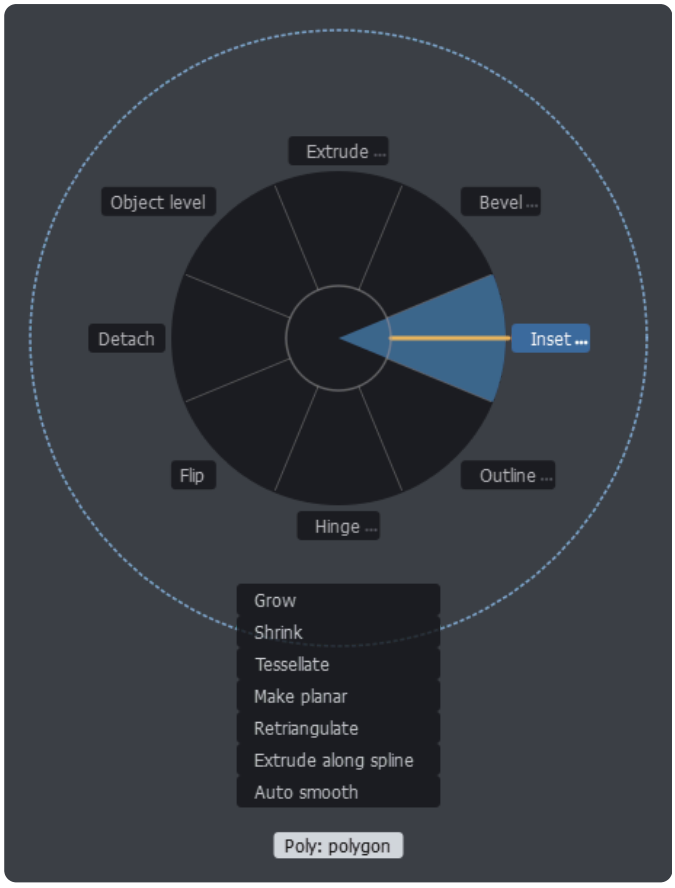
- Let go **near the dial**: no window. The command starts just like its button in the Modify panel. For Inset, Extrude or Bevel that means: drag in the viewport to set the amount.
- Move **further out**, past the **dotted circle**, and let go: the settings window opens. It is the same window as the small settings icon in a quad menu.

The dotted circle **lights up** when you are past it. Move back inside before you let go, and there is no window after all.

A **flick** (lesson 3) works the same way, even with no dial on screen. A short flick runs the command at once. A long one - longer than the way to the dotted circle - opens the window. Your hand learns one movement, fast or slow.



Inside the dotted circle: Inset runs at once.



Past it: the circle lights up - Inset opens its window.

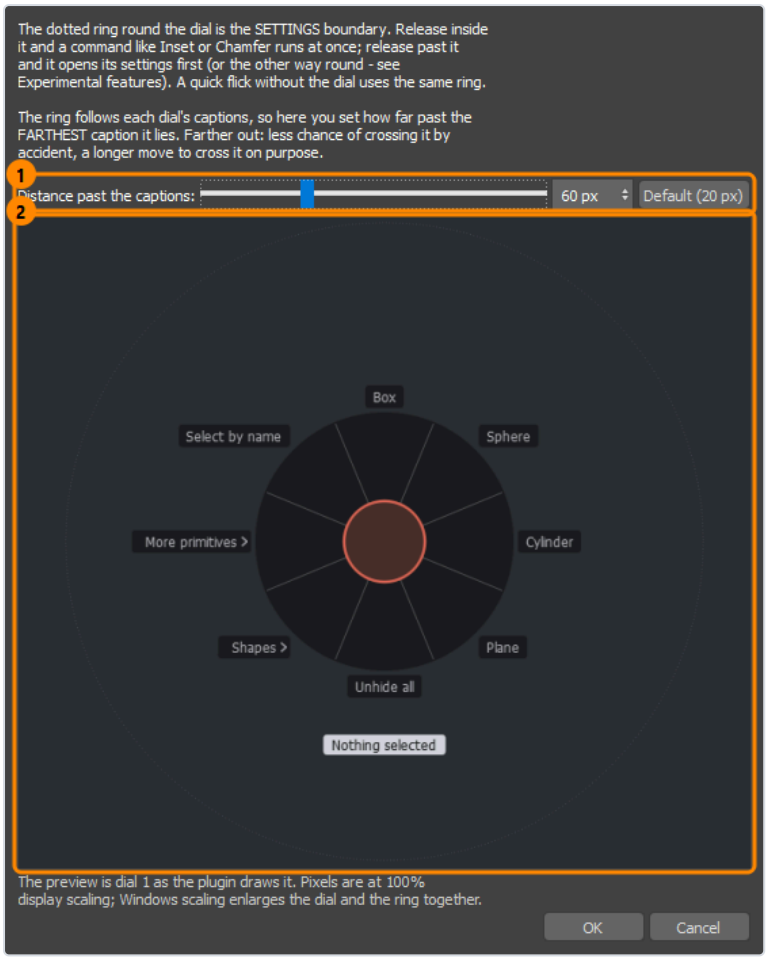
TRY IT

Select a polygon, hold Alt+1, move right (Inset) and let go: Inset starts - drag in the viewport. Do it again, but move far out before letting go: the Inset window opens. Then flick: a short flick right runs Inset at once, a long one opens the window.

Want it the other way round for a command you always set up first - the window unless you go far? Select its direction in the editor and press **Settings...** (lesson 7 shows where).

Is the circle too close or too far? Move it

1. In the editor, press **Settings ring...** at the bottom.
2. Set how far past the captions the circle sits: from 0 to 200 pixels (20 is the normal one). You see the real dial with the circle in its new place.
3. Press OK, then **Save**. Every dial uses this number, and so do flicks.



Settings ring...: how far (1) and the dial with the circle in its new place (2).

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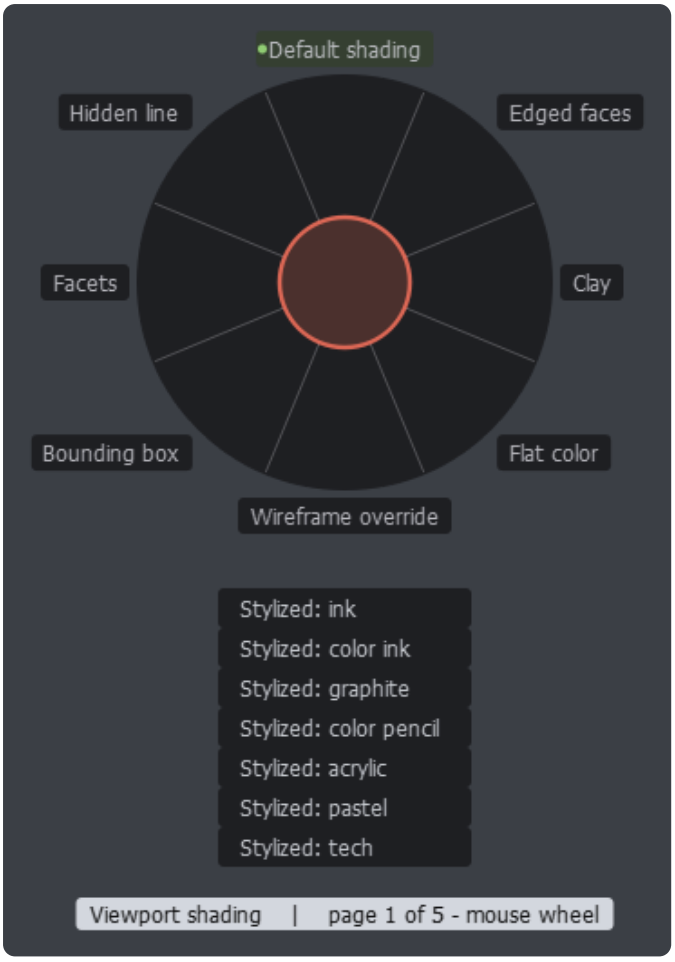
5 Undo by name, repeat, the viewport and the hotbox

4 min

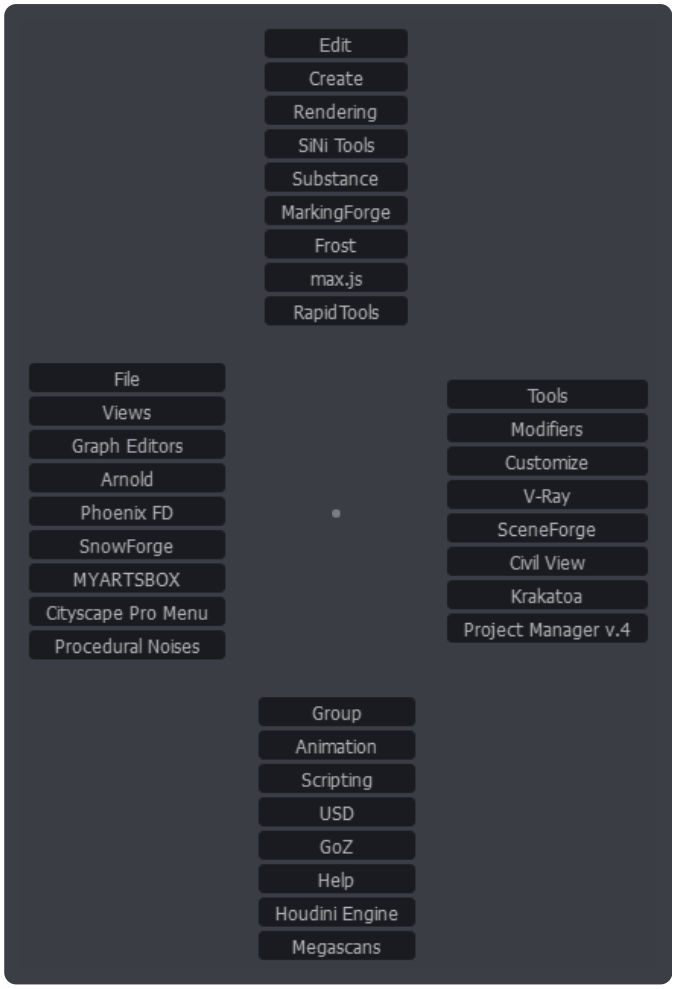
Four dials that save you trips to the menus:

1. **Alt+7** - your last eight steps, by name. Up undoes one step, the next direction (clockwise) undoes two, and so on. Three steps back in one move.
2. **Alt+8** - the commands you ran from the dials, newest first. A flick up repeats the last one.
3. **Alt+5** - how the viewport looks: shading, clay, facets, wireframe and more. Keep holding and turn the **mouse wheel towards you** for more pages: lights and shadows, what is shown (grid, safe frames, statistics, ViewCube...) and the views (top, front, perspective...). A small dot shows a switch that is on. Less common options are in the rows under each page.
4. Hold **Alt+5** and turn the wheel one step **away** from you: the **hotbox**.

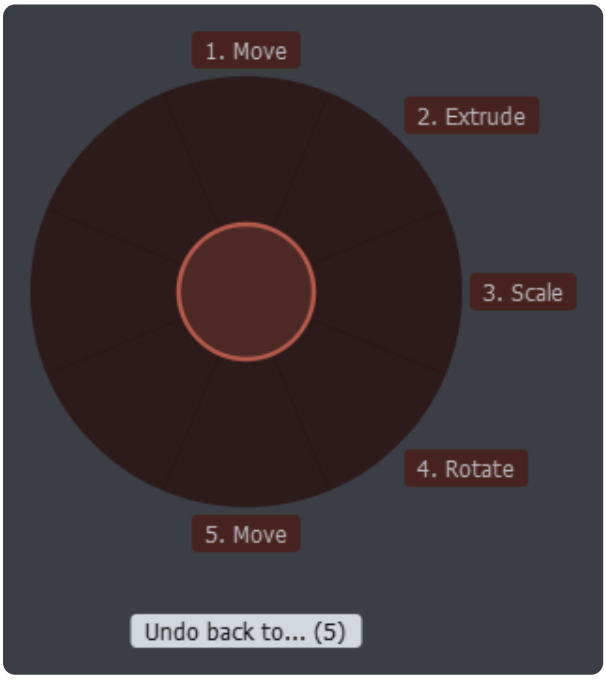
Hotbox - every 3ds Max menu at once, as tiles - menus of other plugins too. Point at a menu title, then at a command, and let go (or click the command). Esc or a right click closes it.



Alt+5 - how the viewport looks. The wheel turns to more pages.



Alt+5, wheel one step away - the hotbox.



Alt+7 - undo by name.

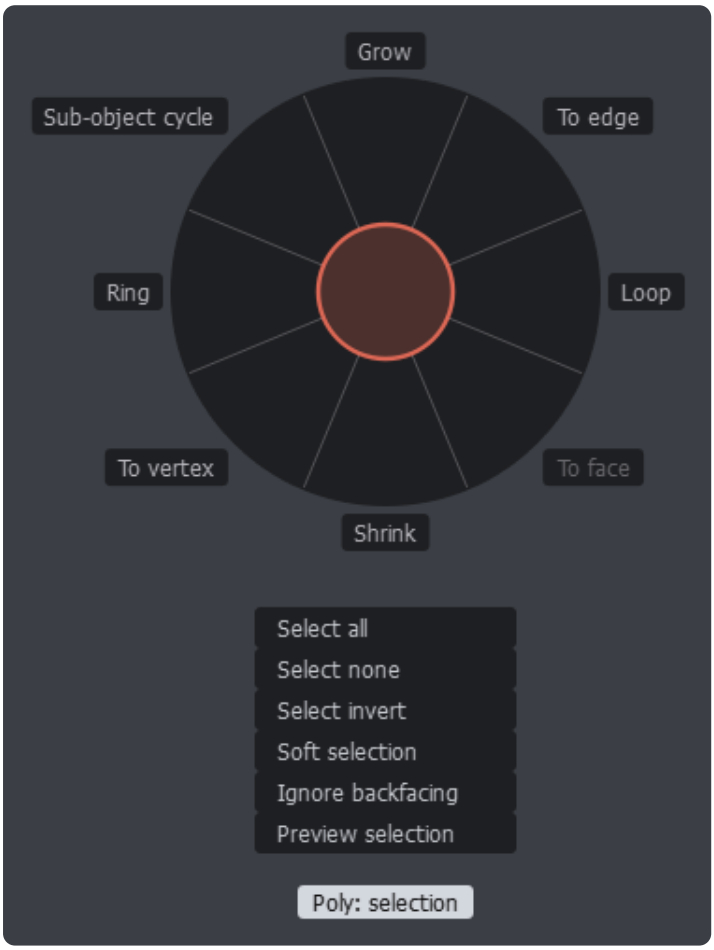
> **Alt+6** fills itself too: it shows the top eight modifiers of the selected object. Pick one to open it in the Modify panel.

6 Selecting and hiding parts of an object

2 min

Other dials follow the level too, not just Alt+1.

1. Work on polygons and hold **Alt+3** (Selection). Grow and Shrink are opposite each other, so are Loop and Ring. On an Editable Poly, three corners turn the selection into vertices, edges or polygons. (On an Edit Poly modifier those corners are Select all, Invert and Select nothing.)
2. At the same level, hold **Ctrl+Alt+6** (Show and hide): it hides or shows polygons, not whole objects.
3. Go back to the whole object (Alt+1, up-left) and hold the same keys: now they work on whole objects.



Alt+3 while working on part of an object.

Five tricks worth knowing

Tricks 2, 3 and 4 work at once. Tricks 1 and 5 must be switched on first in **MarkingForge > Experimental features**. You can switch any of the five off there too.

1. Several commands in one gesture (switch it on first)

1. Switch on **Multiple picks in one gesture** in **MarkingForge > Experimental features**.
2. Hold the key and move out **past the dial**, towards a command.
3. Do not let go. **Come back to the dashed circle** near the middle. The command waits in a queue - a numbered row above the dial (*1 Ring > 2 Connect*).
4. Add more the same way. Then let go inside the dashed circle (it turns green): they all run in order, and **one Ctrl+Z** undoes them all.
5. Changed your mind? Let go in the very middle, or over an empty direction. Nothing runs.



A command waits in the queue (the row above the dial). The dashed circle is where you come back to.

What works in a queue: commands that finish at once - Loop, Ring, Grow, Flip, Connect, Weld, Cap, adding a modifier. Inset, Extrude, Bevel and Chamfer wait for you: the queue stops at such a command - drag in the viewport, then **right-click**, and the next queued command starts by itself. Esc keeps the tool on; another tool, another object or a new gesture drops the rest; each such step is its own Ctrl+Z.

TRY IT

On a polygon (press 4): queue Inset (east) and Extrude (north), let go in the dashed circle. Drag, right-click, drag, right-click.

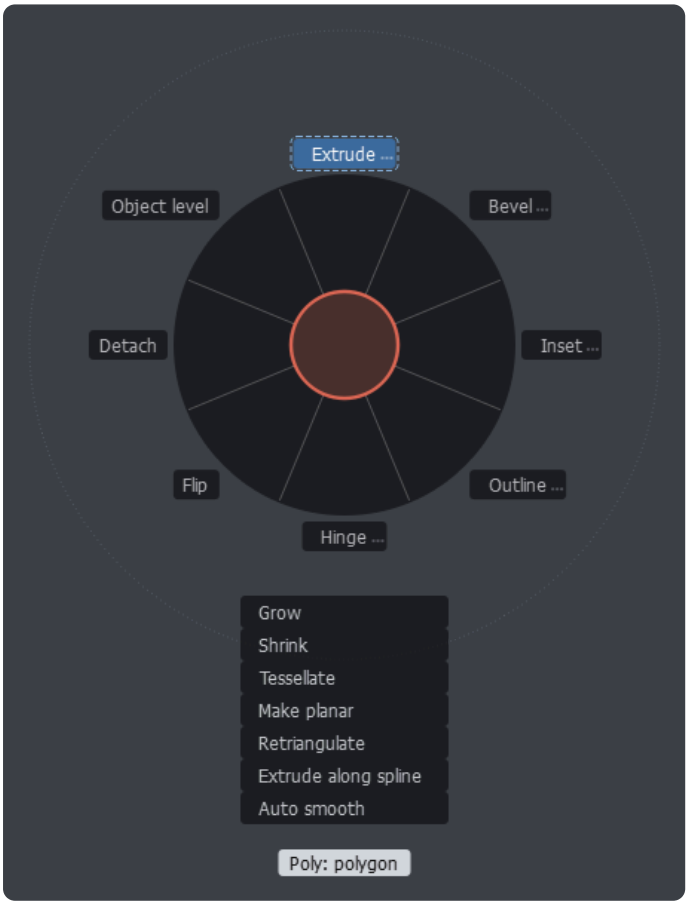
Example: on edges (press 2), select one edge and hold Alt+1 - out to **Ring** and back, out to **Connect** and back, then let go in the dashed circle. A new edge loop goes round. One Ctrl+Z undoes both. A queue you repeat can become one direction: **MarkingForge > Save last chain as an item...**

2. The dial looks at what is under the mouse

The dial looks first at the object **under the mouse**. Only over empty space does it look at what is selected. Point at a line and open Alt+1: you get line tools, even if a box is selected. The command then works on that line - MarkingForge selects it for you, and Ctrl+Z undoes both. If you point at one of several selected objects, they all stay selected. Alt+6 shows the modifiers of the object you point at. A flick uses the object that was under the mouse when you pressed the key. The hotbox and the search never change what is selected.

3. The next command, suggested

MarkingForge notices pairs you repeat - Inset and then Extrude, say. After a pair happens twice, the command you usually use next has an **outline** when the dial opens. A **tap** - press the key and let go without moving - runs it (unless you gave the dial a tap command of its own). It is only a suggestion: aim anywhere else as usual, nothing moves. It forgets when 3ds Max closes.



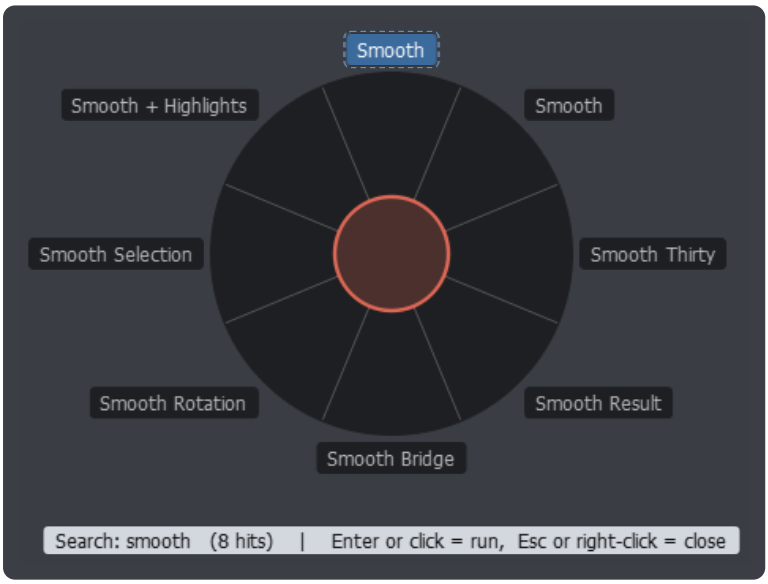
After Inset, Extrude has the outline - a tap runs it.

4. What the marks on the dial mean

You see	It means
A dotted circle around the dial	The settings circle (lesson 4). It lights up when you are past it.
A dashed circle near the middle	A queue is waiting. Come back inside it to add the command (trick 1).
A dashed outline around a direction	That direction is empty. Letting go there cancels.
> after a caption	A submenu. Move through it without letting go - the dial shows what is inside, in the same place (lessons 1 and 13).
The circle in the middle	Let go inside it to cancel - always.
A tile with an outline	What a tap runs - the suggested next command (trick 3).
A small dot before a caption	That switch is on (shadows, grid, edged faces...).
page 2 of 3 - mouse wheel under the dial	The dial has pages. Keep holding the key and turn the mouse wheel (lesson 20).
Nothing selected > More primitives under the dial	You are inside a submenu - the line says which one.

5. Search by typing (switch it on first)

1. Switch on **MarkingForge > Experimental features > Type-to-search**.
2. Hold a dial key and **type letters** - for example `smooth`. The dial turns into a search through every 3ds Max command. It **stays open**, so you can let go of the key.
3. Aim at a result, or press **Enter** for the best one. **Backspace** deletes a letter. **Esc** closes the search.



Alt+1, then `smooth`: the commands with that name, the best one outlined.

> Typing works with the **Alt** and **Shift+Alt** dials. While **Ctrl** is held, letters stay 3ds Max shortcuts (Ctrl+S, Ctrl+Z...), so the Ctrl+Alt dials have no search.

Which key is which

The number (1-8) is the place. **Alt**, **Ctrl+Alt** or **Shift+Alt** is the row. Count from a key you know.

Key	Dial	Key	Dial	Key	Dial
Alt+1	Modelling	**Ctrl+Alt+1**	Transform	**Shift+Alt+1**	Viewport display
Alt+2	Modifiers	**Ctrl+Alt+2**	Align and pivot	**Shift+Alt+2**	Selection sets
Alt+3	Selection	**Ctrl+Alt+3**	Lights and cameras	**Shift+Alt+3**	Materials
Alt+4	Create	**Ctrl+Alt+4**	UV mapping	**Shift+Alt+4**	Render
Alt+5	Viewport shading	**Ctrl+Alt+5**	Snaps and grid	**Shift+Alt+5**	Animation
Alt+6	Modifier stack	**Ctrl+Alt+6**	Show and hide	**Shift+Alt+6**	Scene
Alt+7	Undo	**Ctrl+Alt+7**	View	**Shift+Alt+7**	Tools and setup
Alt+8	Recent commands	**Ctrl+Alt+8**	Viewports	**Shift+Alt+8**	File

Print the cheat sheet and keep it next to the keyboard for the first week: **MarkingForge > Dial cheat sheet (HTML + PDF)...** It shows your dials as small squares with their keys.

MarkingForge - dial cheat sheet

2026-10-07 13:41 · 24 dials and 4 more pages · sorted by shortcut, because that is what the hand presses

Alt+1 Modelling

slot 1

1. when window under the cursor matches slate

↖ NW Pick from selected	↑ N New material	↗ NE Edit selected's material
← W Get from object	Materials (Slate)	→ E Assign to selection
↙ SW Material Explorer	↓ S Material/Map Browser	↘ SE Show in viewport

2. when window under the cursor matches trackView

↖ NW Curve mode	↑ N Tangents: auto	↗ NE Tangents: smooth
← W Isolate curve	Track View	→ E Tangents: linear
↙ SW Frame all	↓ S Flatten	↘ SE Tangents: stepped

3. when window under the cursor matches particleView

↖ NW Rename	↑ N Move up	↗ NE Copy
← W Move left	Particle View	→ E Move right
↙ SW Cut	↓ S Move down	↘ SE Paste

4. when window under the cursor matches sceneExplorer

↖ NW Zoom to selected	↑ N Select all	↗ NE Select children
← W Unhide all	Scene Explorer	→ E Hide
↙ SW Unfreeze all	↓ S Isolate	↘ SE Freeze

List below the dial

1. Expand all
2. Collapse all
3. Find selected
4. Rename

5. when window under the cursor matches commandPanel

Modifier stack. Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.

6. when modifier = Unwrap_UVW

↖ NW Stitch	↑ N Quick relax	↗ NE Pack
← W Weld selected	UV (Unwrap)	→ E Edge loop
↙ SW Break	↓ S Straighten	↘ SE Edge ring

List below the dial

1. Open UV Editor

7. when modifier = Edit_Poly and subobject: 1 - usually vertex

↖ NW Object level	↑ N Chamfer	↗ NE Weld
← W Target Weld	Edit Poly: vertex	→ E Connect
↙ SW Remove	↓ S Extrude	↘ SE Break

List below the dial

1. Grow
2. Shrink
3. Remove isolated

8. when modifier = Edit_Poly and subobject: 2 - usually edge

↖ NW Object level	↑ N Connect	↗ NE Chamfer
← W Remove	Edit Poly: edge	→ E Loop
↙ SW Weld	↓ S Extrude	↘ SE Ring

List below the dial

1. Split
2. Turn
3. Grow
4. Shrink

9. when modifier = Edit_Poly and subobject: 3 - usually border / spline

↖ NW Object level	↑ N Cap	↗ NE Chamfer
← W Grow	Edit Poly: border	→ E Extrude
↙ SW Shrink	↓ S Create shape	↘ SE Connect

List below the dial

1. Bridge

The cheat sheet, ready to print.

PART 2

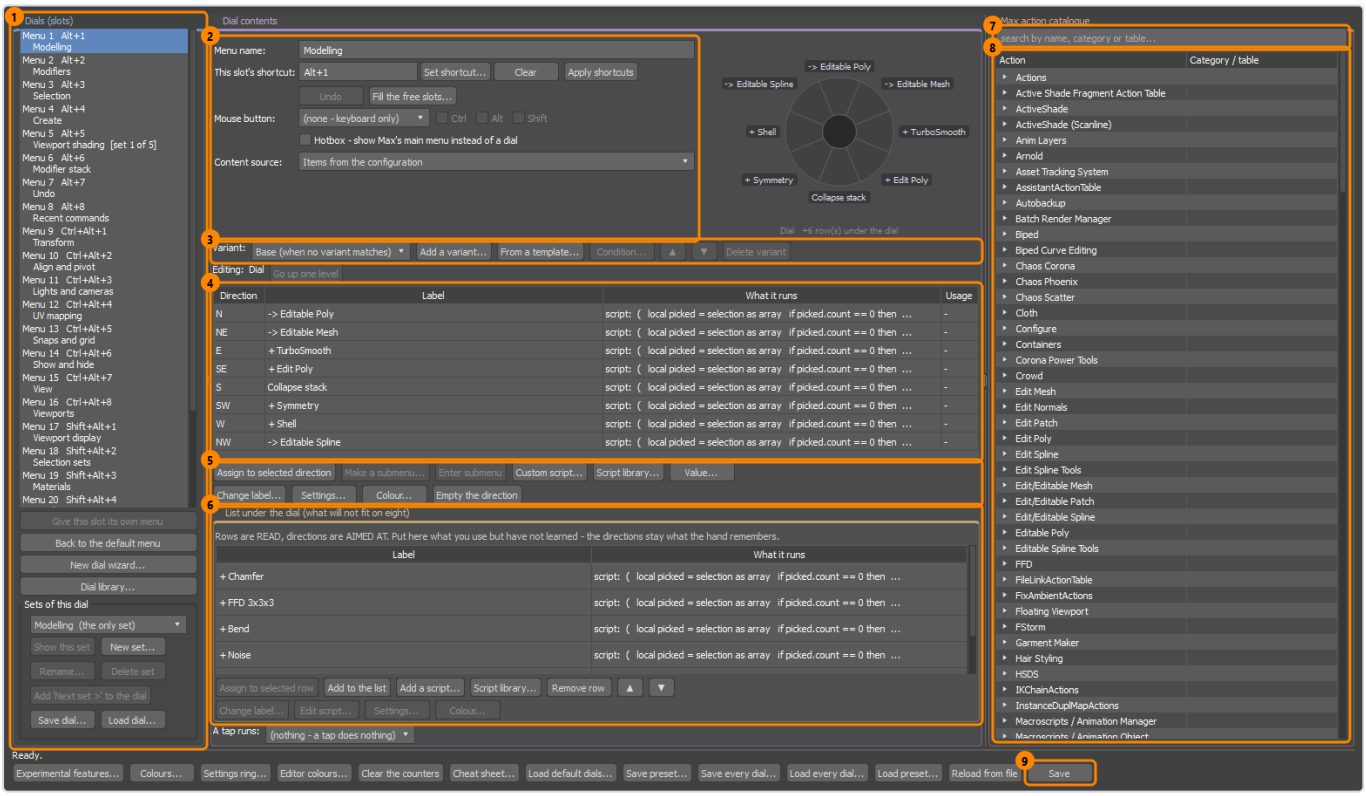
Changing the dials

Sixteen lessons in the menu editor. Nothing reaches your dials until you press **Save**, so try anything you like. **Reload from file** throws away what you have not saved. One exception: the colours of the editor window itself (lesson 17) are kept as soon as you press OK.

7 What is where in the editor

2 min

Open **MarkingForge > Menu editor...**



The menu editor.

1. **Dials** - all 24 dials and their keys. Click one to change it. Right-click it to save it to a file or load one.
2. **Dial contents** - the dial's name, key, mouse button, and where its commands come from.
3. **Variant** - other versions of the dial for other situations (lesson 12).
4. **Directions** - the eight directions, N to NW: the caption, what runs, how often you used it.
5. **Direction tools** - buttons for the selected direction: submenu, script, script library, value, caption, settings, colour, empty.
6. **List under the dial** - the extra rows.

7. **Search** - find any 3ds Max command by its name.
8. **Catalogue** - every 3ds Max command (over 4000), in groups.
9. **Save** - saves your dials. 3ds Max uses them at once.

Next to the dial contents, the editor **draws the dial** you are changing. Click a part of the drawing to select that direction. Under the list of dials are **New dial wizard...**, **Dial library...** and **Sets of this dial**. A dial with several pages shows them as **tabs** above its contents (lessons 20-22). The bar at the bottom is for all dials at once: colours, the cheat sheet, presets, and saving or loading every dial (lessons 16-18).

> **You cannot lose your work by accident.** If something is not saved, Esc or the window's X asks you: **Save**, **Discard** or **Cancel**.

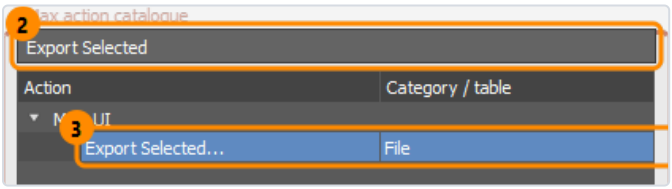
---

8 Put a command in a direction

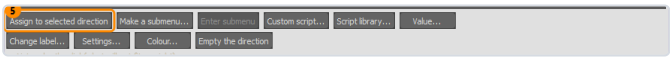
3 min

This is the one skill everything else builds on. Example: on the **File** dial (Shift+Alt+8), swap *Save copy as* (down-right, SE) for **Export Selected**.

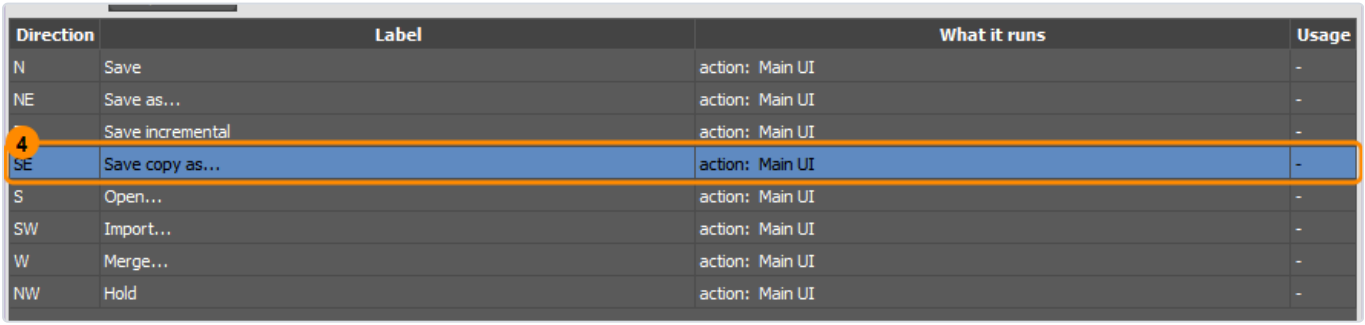
1. In the list of dials, click **Menu 24** (Shift+Alt+8, File).
2. Type `Export Selected` in the search box.
3. Click **Export Selected...** in the catalogue.
4. Click the **SE** row in the directions table.
5. Press **Assign to selected direction** under the table. (Or double-click the command, or drag it onto the row.)
6. Press **Save**.



Steps 2 and 3: type, then click the command.



Step 5: the button under the directions table.



Step 4: the SE row is selected - it still holds Save copy as.

Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
E	Save incremental	action: Main UI	-
SE	Export Selected...	action: Main UI (name from Max)	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

After step 5: SE runs Export Selected. *(name from Max)* means the caption is 3ds Max's own name - lesson 9 makes it shorter.

TRY IT

Hold Shift+Alt+8 over a viewport: down-right is now Export Selected.

The same command can be in several groups (Select All is in seven). The *Category* column tells them apart. For commands that work on the scene, pick *Main UI*.

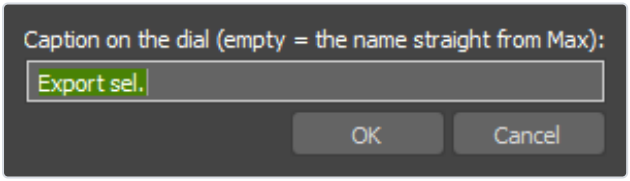
---

9 A shorter caption

1 min

Short captions keep the dial easy to read.

1. Select the direction.
2. Press **Change label...**
3. Type the caption and press OK. Leave it empty to use 3ds Max's own name.
4. Save.



Two words are better than a cut-off name.

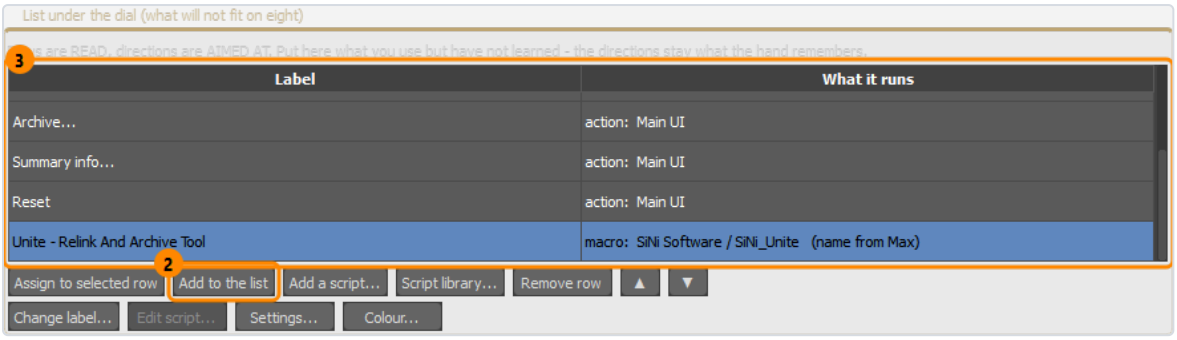
---

10 Rows under the dial

2 min

Rows let one key hold more than eight commands.

1. Click the dial.
2. Find the command in the catalogue.
3. Press **Add to the list** (or drag the command into the list).
4. Change the order with ▲ ▼. Remove a row with **Remove row**. Up to twelve rows.
5. To change a row, select it. The list has its own buttons: **Change label...**, **Edit script...**, **Settings...** and **Colour...**. **Add a script...** adds a row that runs a script of your own.
6. Save.



The rows, and Add to the list under them.

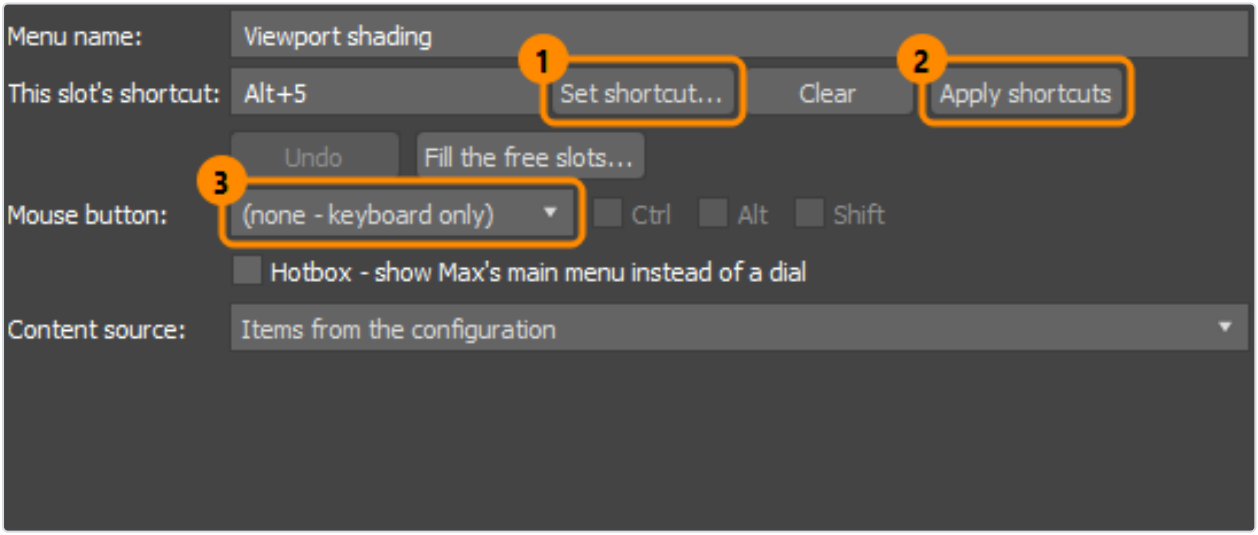
> Put here what you use but your hand does not know yet. When your hand wants it, move it to a direction.

---

11 Change a dial's key, or open it with the mouse

2 min

1. Click the dial.
2. Click **Set shortcut...** A small window opens. Press the new keys and click **Set**. If the key already opens another dial, that is fine - it moves to this one. **Clear** removes the key.
3. Press **Apply shortcuts**. The keys go into 3ds Max. Your other shortcuts stay as they are, and a backup is made. **Undo** next to it brings back the shortcuts from before. **Fill the free slots...** gives the usual key to every dial that has none.
4. If you like, pick a **Mouse button** too, together with at least one of Ctrl / Alt / Shift - for example Ctrl+Shift + *middle button*.
5. Save.



Set shortcut (1), Apply (2), mouse button (3).

You also find the keys in 3ds Max's **Customize > Hotkey Editor**, as *MarkingForge: Menu 1 ... 24*.

---

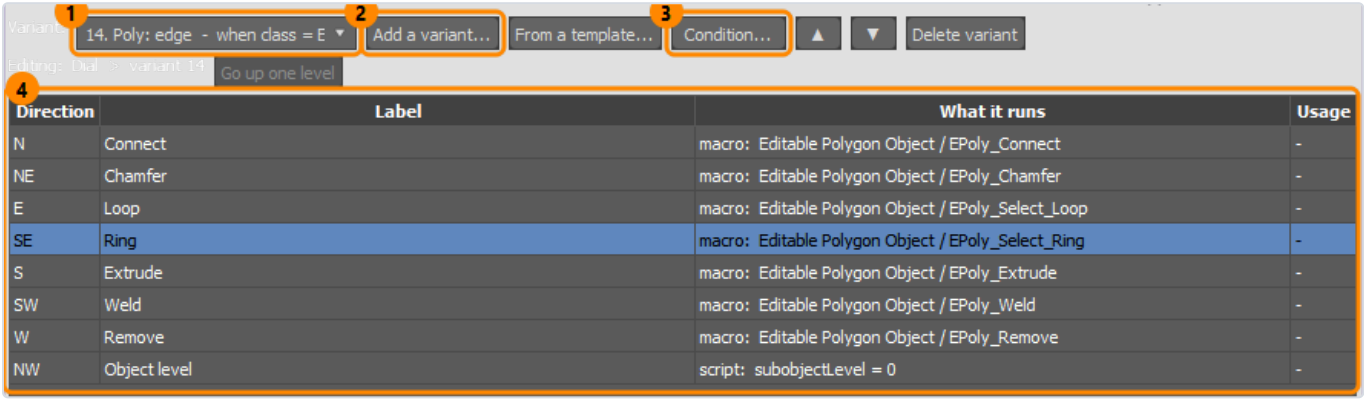
12 Same key, other commands in another situation

4 min

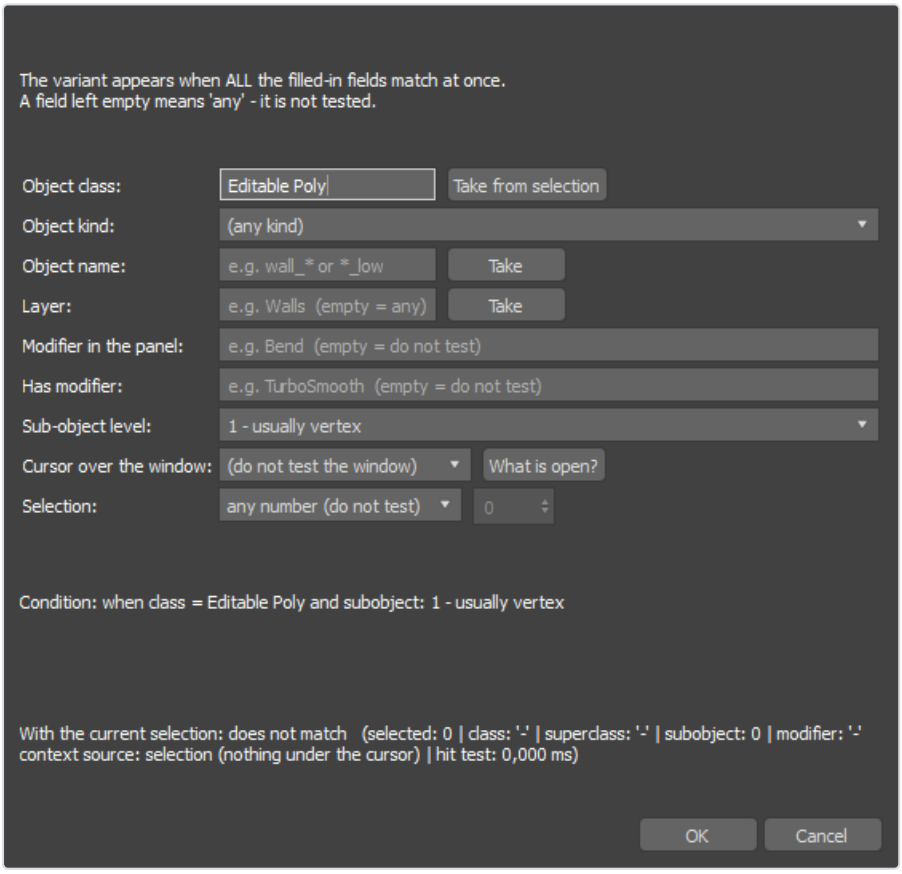
Variant - another version of a dial. It shows only in a situation you choose - for example when you work on the edges of an Editable Poly. The built-in Alt+1 uses variants to show vertex tools on vertices and edge tools on edges.

Example: give a dial other commands when you work on the edges of an Editable Poly.

1. Click the dial and press **Add a variant...** It starts as a copy of what you see.
2. A window opens. In 3ds Max, select an Editable Poly, go to edges (key 2) and press **Take from selection**. The object type and the level fill in by themselves.
3. Press OK. The **Variant** list now shows your variant. Fill its directions like any dial.
4. If two variants fit at the same time, the higher one in the list wins. Keep the more exact ones on top (▲ ▼).
5. Save.



Alt+1's built-in edge variant: the list of variants (1), Add a variant (2), the situation it needs (3) and its own directions (4).



The situation window (here: Editable Poly, vertices). Empty fields are not checked.

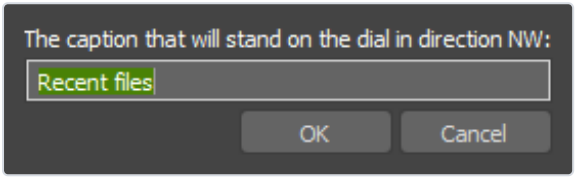
> A variant can depend on more than the object. It can also depend on: the **window under the mouse** (Slate, Track View, Scene Explorer...), the modifier open in the Modify panel, how many objects are selected, an object's name, or its layer.

13 A submenu

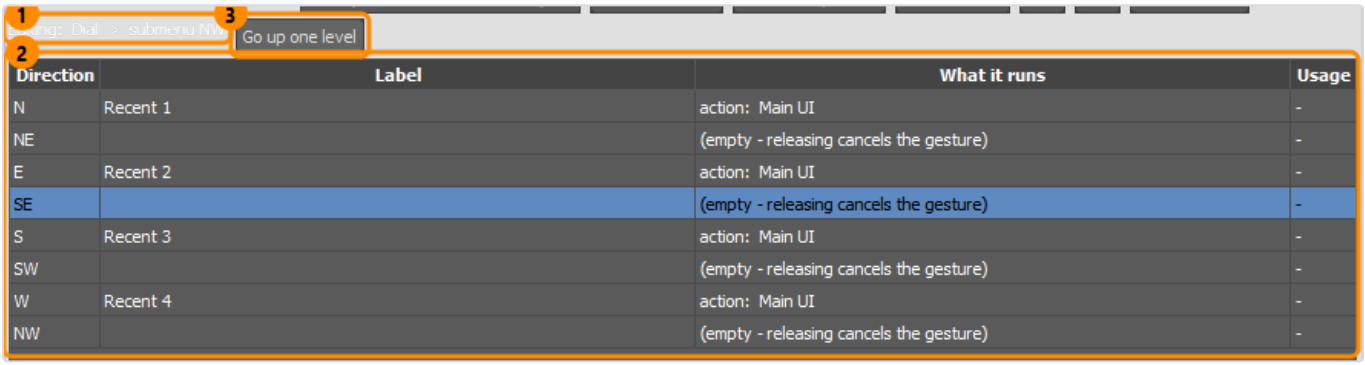
3 min

A submenu puts a whole dial behind one direction (see lesson 1).

1. Select a direction and press **Make a submenu...**
2. Type its caption - it is what the dial shows.
3. You are now inside the submenu. The line above says *Editing: Dial > submenu NW*. Fill its directions.
4. Press **Go up one level** to go back. Save.



Step 2: the caption.



Direction	Label	What it runs	Usage
N	Recent 1	action: Main UI	-
NE		(empty - releasing cancels the gesture)	-
E	Recent 2	action: Main UI	-
SE		(empty - releasing cancels the gesture)	-
S	Recent 3	action: Main UI	-
SW		(empty - releasing cancels the gesture)	-
W	Recent 4	action: Main UI	-
NW		(empty - releasing cancels the gesture)	-

Step 3: inside the submenu - where you are (1), its own directions (2), the way back (3).

TRY IT

On the dial, move through the submenu's direction without letting go. The dial stays in place and shows the submenu. The line under it says where you are (*Nothing selected* > *More primitives*). Move back inside the dial to pick. Move into the middle to go back.

Submenus in the dials that come with MarkingForge. Open them to see what submenus are good for - groups too big for eight directions:

Dial	Direction	Submenu
Alt+4 Create	S, SW, W, NW	Shapes, More primitives, Helpers, Cameras and lights
Alt+2 Modifiers	SW, NW	Deform (Bend, Twist, Taper, FFD...), Geometry (Mirror, Array, Cap Holes, OpenSubdiv...)
Alt+1 Modelling, nothing selected	SW, W	Shapes, More primitives

In the editor's drawing, a submenu is marked with >. **Double-click** it (or its row in the table) to go inside.

14 Scripts on a dial - ready-made or your own

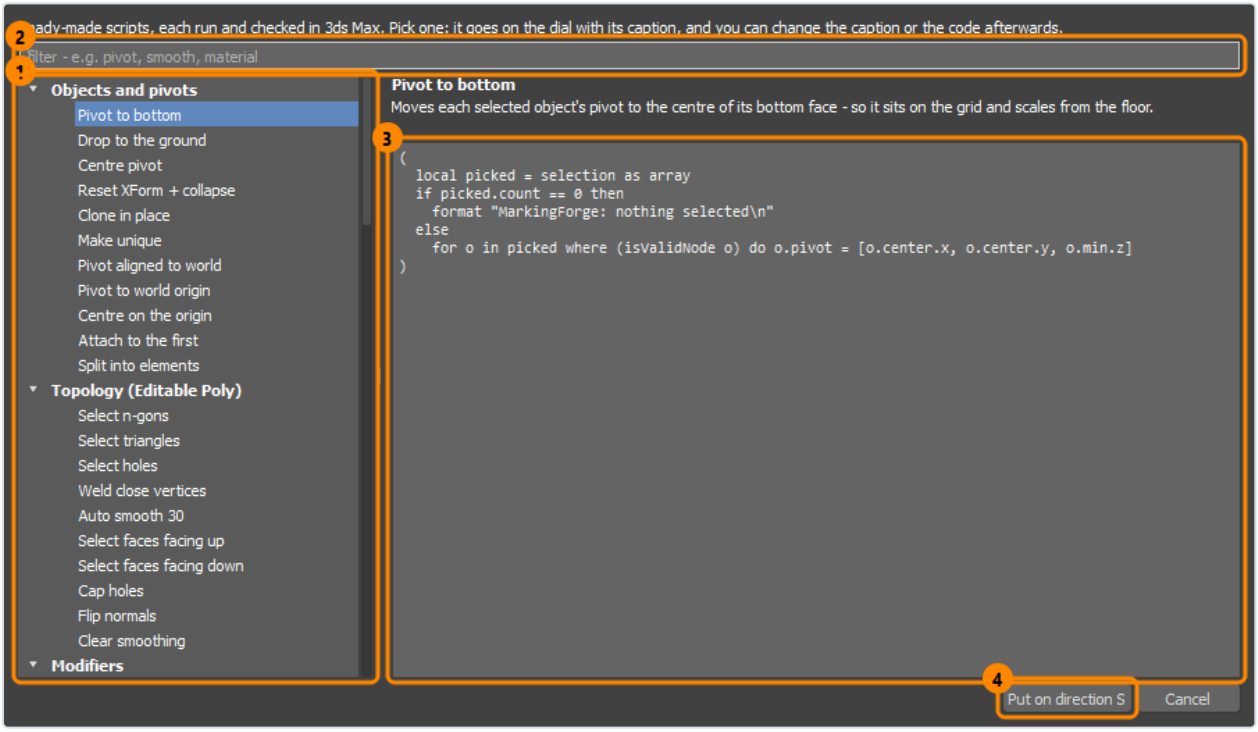
6 min

Script - a small program for 3ds Max, written in its language, MAXScript. You do not have to write one - 102 ready-made scripts come with MarkingForge.

The **Script library** has its 102 scripts in eleven groups: objects and pivots, topology, modifiers, advanced modelling (loops and rings, chamfer, bridge, bevel, planarize, symmetrize, greeble...), UV mapping, materials and display, selection and moving, layout and placing, shapes and lines, scene and layers, cameras, lights and render. Each one was tried in 3ds Max before it went in.

A. From the library - nothing to write

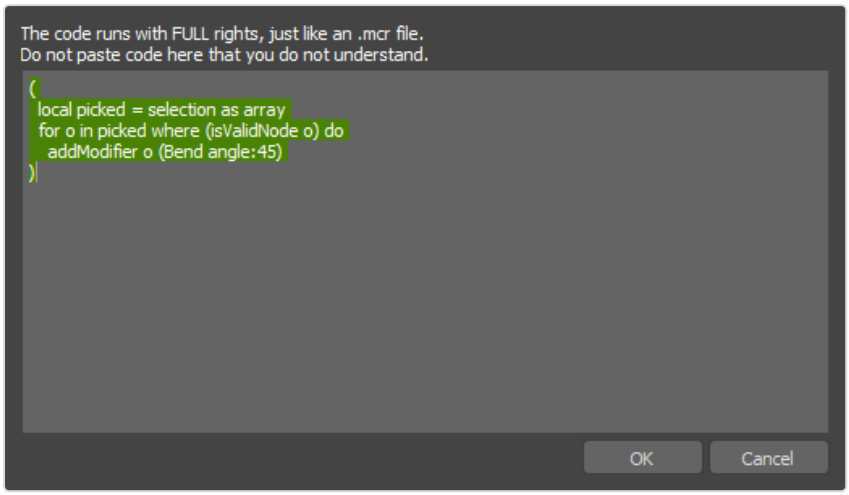
1. Click the dial and select a direction. (To keep your directions, use the list under the dial.)
2. Press **Script library...** (under the directions, or in the list's buttons).
3. Click a script on the left. The right side says what it does, and shows its code if you want to look.
4. To find one, type in the filter: *pivot*, *smooth*, *material*.
5. Press **Put on direction S** - the button names the direction you selected - or **Add to the list**. Then **Save**.



The script library: groups on the left, what the script does and its code on the right.

B. A script of your own

1. Select a direction and press **Custom script...** (once the direction holds a script, the button says **Edit script...**).
2. Paste the script and press OK.
3. Give it a caption with **Change label...**
4. Save.



A script that adds a Bend to every selected object.

Four scripts every modeller uses

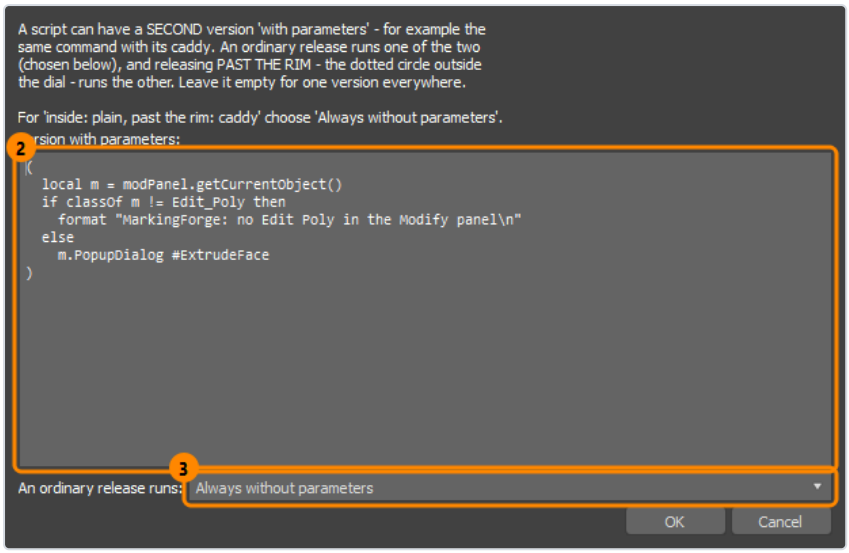
All four are in the library. Put them in the list under Alt+1 (**Menu 1**) and they are one gesture away, whatever you model.

Script	Try it like this
Pivot to bottom	Make a sphere, hold Alt+1, pick the row: the pivot jumps to the bottom of the sphere. With **Drop to the ground** after it, objects stand on the grid.
Select n-gons	Turn a cylinder into an Editable Poly, pick the row: both caps are selected (polygons with more than four sides), and the Listener window says how many. Fix them before smoothing.
Reset XForm + collapse	Scale a box to 300 % and turn it, then pick the row: the size and turn become part of the mesh, and the object is clean - what Shell, Symmetry and exporters expect.
Smoothing on/off	Give a few objects TurboSmooth, select them all, pick the row: every TurboSmooth switches off at once. Pick it again and they come back.

1. Click **Menu 1** on the left of the editor.
2. In the list's buttons press **Script library...**, pick **Pivot to bottom**, press **Add to the list**.
3. Do the same for the other three.
4. Save, then hold Alt+1: the four rows are under the dial.

C. One script in two versions - plain, and with a settings window

1. Select a direction that runs a script and press **Settings...**
2. Paste the second version into **Version with parameters**. For Extrude on an Edit Poly modifier: `(modPanel.getCurrentObject()).PopupDialog #ExtrudeFace`
3. Choose **Always without parameters**. Now letting go near the dial runs the plain version, and letting go past the dotted circle runs the version with the settings window.
4. OK, then Save. In the table, the script now says *[+ with parameters]*.



Settings... on a script: the version with parameters, and which version a normal release runs.

> The library's **Extrude (Edit Poly)** already works this way. A script can do anything 3ds Max can - only use scripts you trust. The Reference (chapter 8) lists every library script.

---

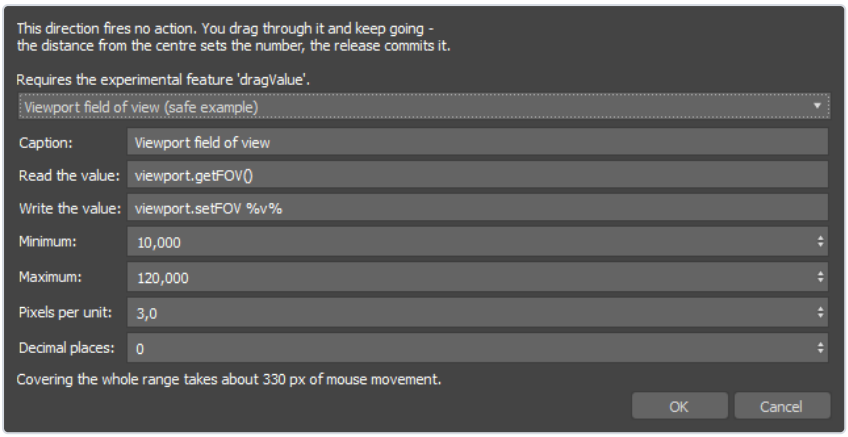
15 Values - five gesture sliders

10 min

> **Value** - a direction that works like a slider. Hold the key, move into that direction and keep going. The further you go, the bigger the number - you see it next to the caption. Let go, and the number is set in the scene. Moving back makes it smaller.

You do not have to write anything. The **Value...** window has ready-made examples: pick one and every field fills in. The five below are the first five in that list - the tables only show you what you will see. The two code fields just say where the number is read from and where it goes (`%v%` is the number).

1. Switch on **MarkingForge > Experimental features > Direction that sets a value** (in the editor: **Experimental features...** in the bottom bar). Without it, the direction does nothing, and the dial says so.
2. In the editor, click a dial, select an **empty direction** and press **Value...**
3. Pick an example from the list at the top of the window. Every field fills in. Change the caption if you like.
4. Look at the line at the bottom of the window. It says how far you move the mouse from the smallest number to the largest. About 150 to 500 pixels feels right. **Pixels per unit** (how far you move for one step) changes it.
5. **OK**, then **Save**. The dial in 3ds Max uses it at once.



The Value window with the first example, the viewport field of view.

Example 1 - Viewport field of view

Zooms the lens of the perspective view. It needs nothing selected and changes only the view - the safest one to start with.

Field	Value
Caption	Viewport field of view
Read the value	`viewport.getFOV()`
Write the value	`viewport.setFOV %v%`
Minimum / Maximum	10 / 120
Pixels per unit	3
Decimal places	0

TRY IT

In a perspective view drag into the direction and on: the view widens as you move out, narrows as you come back; release at the angle you like.

Example 2 - Time slider frame

Moves the time slider. Three pixels of movement = one frame.

Field	Value
Caption	Time slider frame
Read the value	`sliderTime.frame`
Write the value	`sliderTime = %v%`
Minimum / Maximum	0 / 100
Pixels per unit	3
Decimal places	0

TRY IT

Move out slowly and watch the time slider follow; release on the frame you want.

Example 3 - TurboSmooth iterations of the selection

Sets how much TurboSmooth smooths every selected object. 40 pixels per step, so each step is easy to hit.

Field	Value
Caption	TurboSmooth iterations of the selection
Read the value	`selection[1].modifiers[#TurboSmooth].iterations`
Write the value	`for o in selection where o.modifiers[#TurboSmooth] != undefined do o.modifiers[#TurboSmooth].iterations = %v%`
Minimum / Maximum	0 / 4
Pixels per unit	40
Decimal places	0

TRY IT

Select a few objects with TurboSmooth; the number next to the caption steps 0, 1, 2, 3, 4 as you move.

Example 4 - Uniform scale of the selection

Makes every selected object bigger or smaller, evenly - from a tenth to five times.

Field	Value
Caption	Uniform scale of the selection
Read the value	`selection[1].scale.x`
Write the value	`for o in selection do o.scale = [%v%, %v%, %v%]`
Minimum / Maximum	0.1 / 5
Pixels per unit	60
Decimal places	2

TRY IT

Select objects, move out: they grow in the viewport only after the release - the number shows where you are while you move.

Example 5 - Thickness of the selected splines

Turns the selected lines and circles into tubes and sets how thick they are.

Field	Value
Caption	Thickness of the selected splines
Read the value	`selection[1].render_thickness`
Write the value	`for o in selection where superClassOf o == shape do (o.render_renderable = true; o.render_displayRenderMesh = true; o.render_thickness = %v%)`

Minimum / Maximum	0.1 / 20
Pixels per unit	15
Decimal places	1

TRY IT

Draw a few lines or circles, select them and set the thickness in one move.

? next to the caption means there is nothing to read - usually nothing is selected, or the wrong kind of object. Letting go then changes nothing. Examples 3-5 change **every** selected object and skip the ones they do not fit (no TurboSmooth, not a line).

Changed your mind? Move into the middle of the dial and let go - nothing changes.

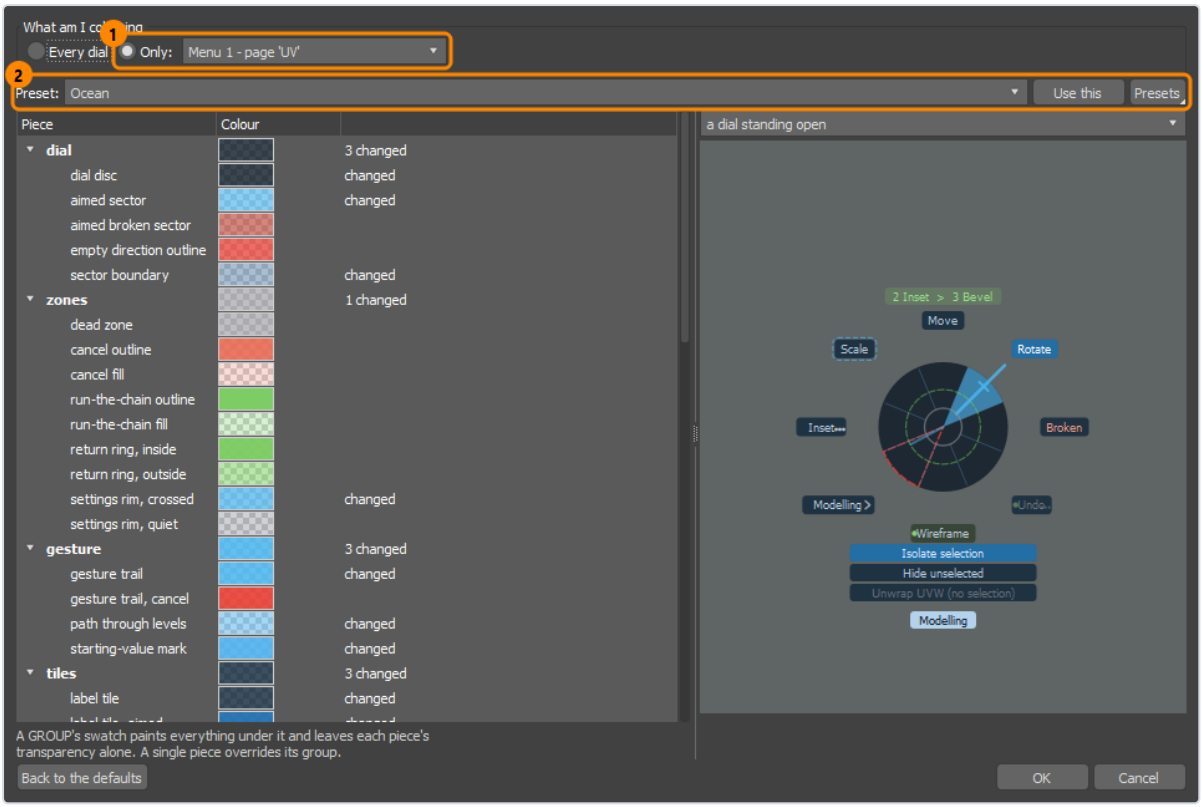
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16 Colours and ready-made looks

3 min

A colour helps you see which dial opened before you read a word.

1. Press **Colours...** in the bottom bar of the editor.
2. Under **What am I colouring**, choose **Every dial** or **Only** this one. A dial can have pages (lesson 20). Then the **Only:** list also offers *Menu N - every page*, and each page on its own (*Menu N - page 'UV'*).
3. Pick a look under **Preset:** and press **Use this**. There are thirty ready-made looks - Graphite, Ocean, Ember, Neon, High contrast and more, five of them with light tiles - and the *MarkingForge default*.
4. Or click a group (dial, zones, tiles, text, list...) or a single piece and pick a colour. The list above the preview shows the moment you are colouring: the dial open, aiming at a direction, about to cancel, a queue waiting.
5. OK, then Save.



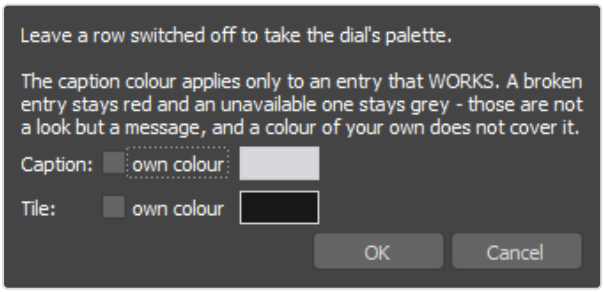
Colours...: what you colour - here one page of Menu 1 (1) - and the ready-made looks (2).

> **Your own looks. Presets > Save these colours as a preset...** keeps your look under a name. **Export this preset to a file...** saves it, to take to another computer. **Import a preset from a file...** brings it in. A dial that uses the default menu has no colours of its own - press **Give this slot its own menu** first.



The thirty ready-made looks.

For one command only - a red caption on a command that deletes things, say - select its direction and press **Colour...**:



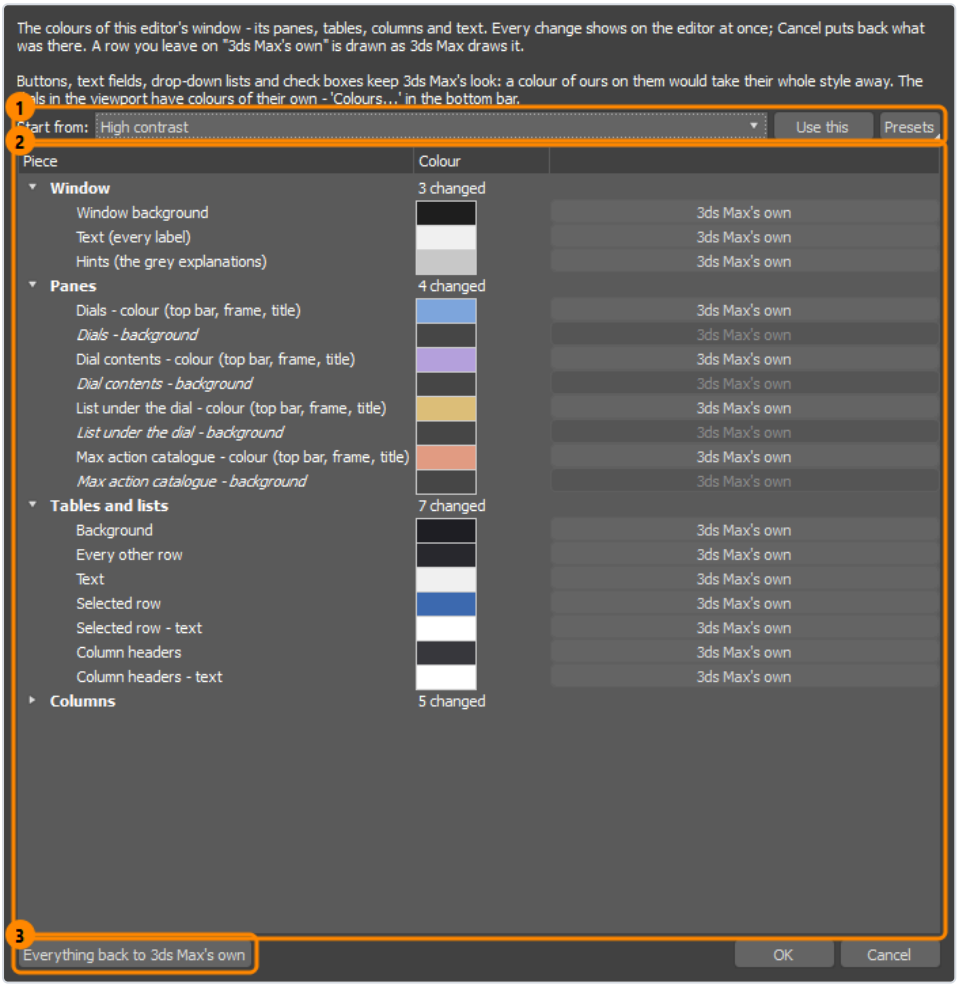
The caption and tile colour of one command.

17 The colours of the editor window

2 min

This colours the editor window itself - not the dials.

1. Press **Editor colours...** in the bottom bar.
2. Pick a starting point under **Start from:** and press **Use this**. There are 21 ready-made looks (High contrast, Calm grey, Nord, Espresso...) and *3ds Max's own*.
3. Change any piece: the window, each pane, the tables and lists, each column on its own. The editor changes as you pick. **3ds Max's own** on a row removes that colour.
4. OK keeps it - no Save needed. Cancel puts back what was there.



Editor colours...: the starting points (1), every piece of the window (2), everything back to 3ds Max's look (3).

> Buttons, text fields, lists and tick boxes keep 3ds Max's look. **Presets** saves, exports and imports these colours, as in lesson 16.

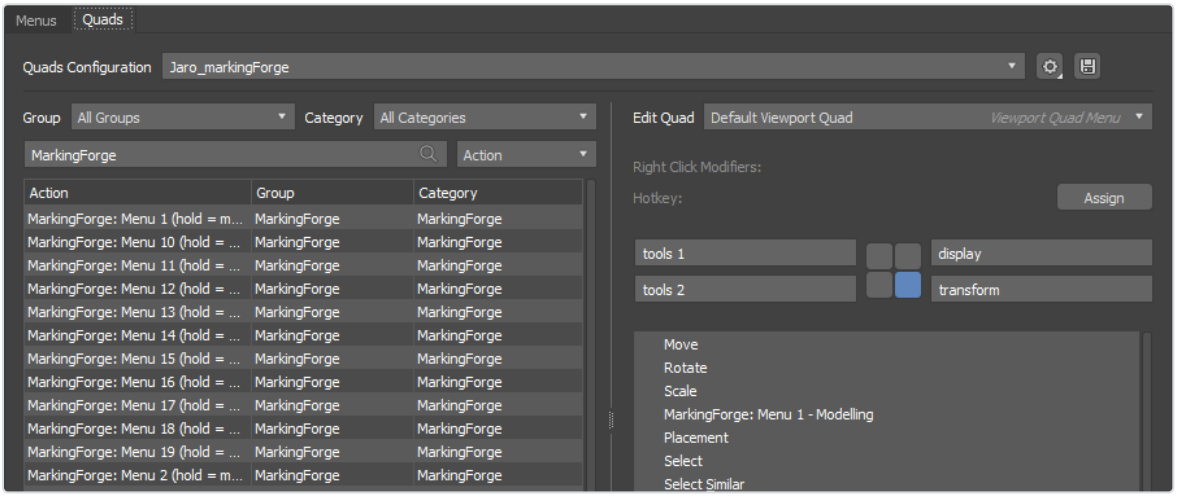
18 Print, share, start again - and quad menus

3 min



Cheat sheet (1), Load default dials (2), Save preset (3), Load preset (4), Save (5), Save every dial (6), Load every dial (7).

1. **Cheat sheet...** - a PDF of your saved dials to print, with keys and mouse buttons. Save first: changes not saved are left out.
2. **Save preset...** - your whole layout in one file, to share: the dials, their pages and their colours. **Load preset...** shows what a file holds before loading it, and by default removes the scripts from other people's presets.
3. **Load default dials...** - the dials MarkingForge came with. By default only for the dials you have not changed. Tick *Replace...* to get all of them back as they came, pages and colours too.
4. **Save every dial...** - each of the 24 dials and each of their pages as its own file, in a new folder named with the date. **Load every dial...** puts such a folder back - every dial, its pages and its colours. Scripts in the files are left out unless you choose to keep them, and nothing changes until you Save.
5. Any MarkingForge command can go into a right-click quad menu: **Customize > Menu Editor > Quads**, search `MarkingForge`, drag a command into a quad, **Done**.



Menu Editor, Quads tab, filtered to MarkingForge.

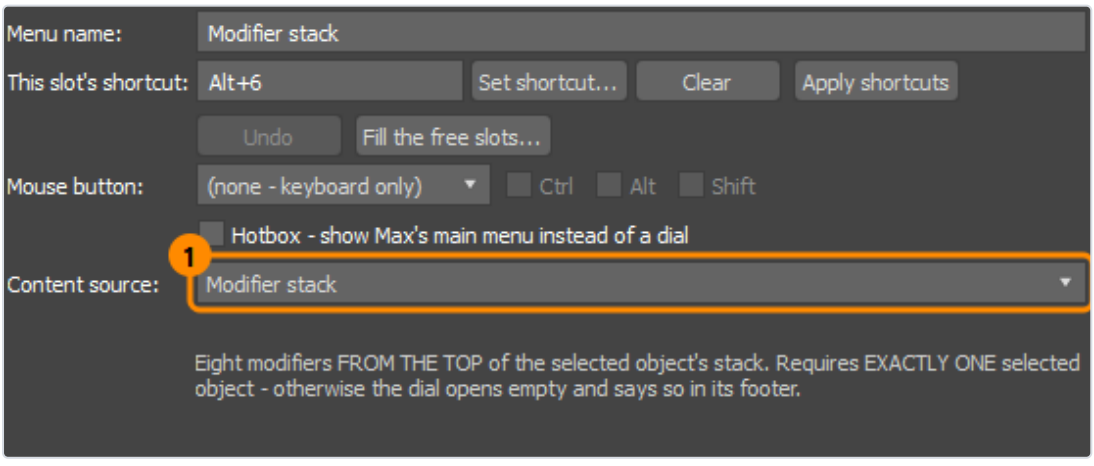
---

19 Dials that fill themselves

6 min

Content source - a setting under *Dial contents*. It decides where a dial's eight directions come from. Normally (*Items from the configuration*) they are the commands you put there. The other choices **fill the dial by themselves** each time it opens - from the scene or from what you just did.

The directions you had typed are kept, and come back if you switch back.



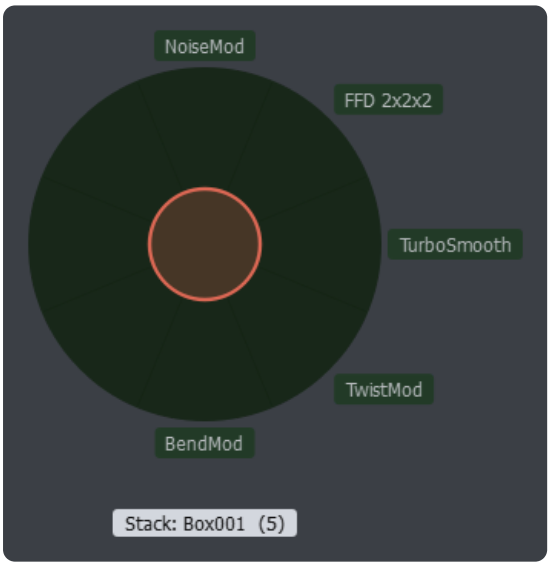
Content source, under the shortcut and the mouse button.

Choice	The directions are	Letting go on one	Dial that does it
Items from the configuration	what you put there	runs it	most dials
Modifier stack	the top eight modifiers of the ONE selected object	opens that modifier in the Modify panel	Alt+6
Undo stack	your last eight steps, by name	undoes back to it - up one step, the next direction two, and so on	Alt+7
Recent commands	the eight commands you ran last, newest first (from any dial, a tap or the search)	runs it again	Alt+8
Selection sets	the scene's named selection sets	selects that set	Shift+Alt+2

Undo stack, Recent commands and Selection sets each have a switch in **Experimental features**; all three are on from the start. The **Hotbox** tick box above Content source is one more way to fill a dial: every 3ds Max menu as tiles (the last page of Alt+5).

Modifier stack - step by step

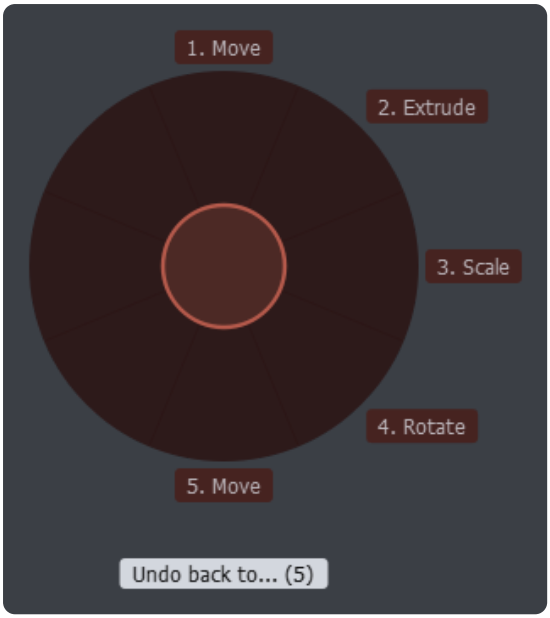
1. Pick a dial to turn into a modifier dial (or give a free slot its own menu).
2. Set **Content source** to **Modifier stack** and press **Save**.
3. In the scene, select ONE object with a few modifiers - a box with Bend, Twist and TurboSmooth.
4. Hold the dial's key: the modifiers are in the directions, the top one first. Let go on Bend - the Modify panel jumps to Bend.



Alt+6 - the modifiers of the selected box.

Undo stack - step by step

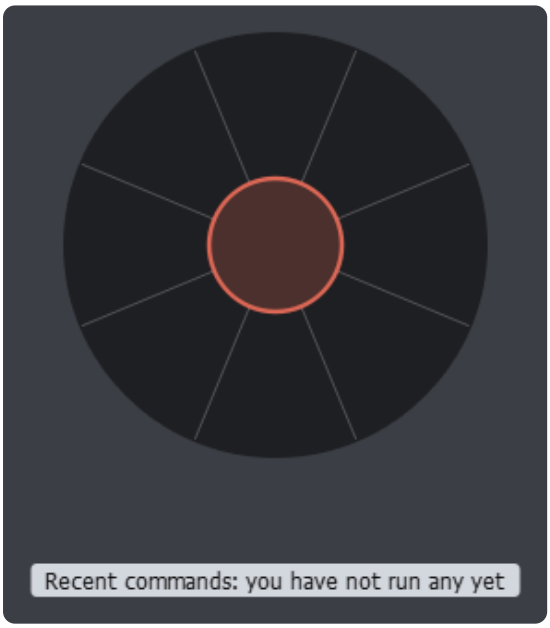
1. On a dial, set **Content source** to **Undo stack** and Save.
2. Do a few things: move, extrude, add a modifier.
3. Hold the key: your steps are there by name, the last one up. Let go three directions round - three steps undone in one move.



Alt+7 - undo by name.

Recent commands - step by step

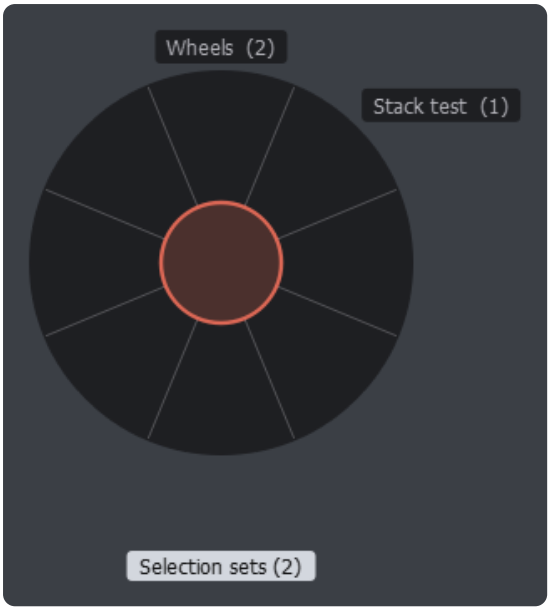
1. Set **Content source** to **Recent commands** and Save.
2. Run a few commands from any dial (a flick, a tap, a row or the search).
3. Hold the key: they are there, newest up. A flick up repeats the last one. The list starts empty each time 3ds Max starts.



Alt+8 - recent commands.

Selection sets - step by step

1. Select a few objects. In the main toolbar's **Named Selection Sets** box, type a name and press Enter. Make two or three sets.
2. Set **Content source** to **Selection sets** and Save.
3. Hold the key: each set is a direction, with how many objects it has. Let go on one - it is selected.



Shift+Alt+2 - the scene's selection sets.

Only one variant of a dial filled this way

1. On a dial press **Add a variant...** In the window that opens, set **Cursor over the window** to **Command panel** and press OK.
2. With the new variant selected, set **Content source** to **Modifier stack**.
3. Save. The same key now shows your commands over a viewport, and the modifiers over the command panel. The built-in Alt+1 does exactly this.

> If such a dial has nothing to show, it opens empty. That is not an error. The line under it says why: *Modifier stack: select ONE object*, *Undo: nothing to undo*, *Selection sets: the scene has none*.

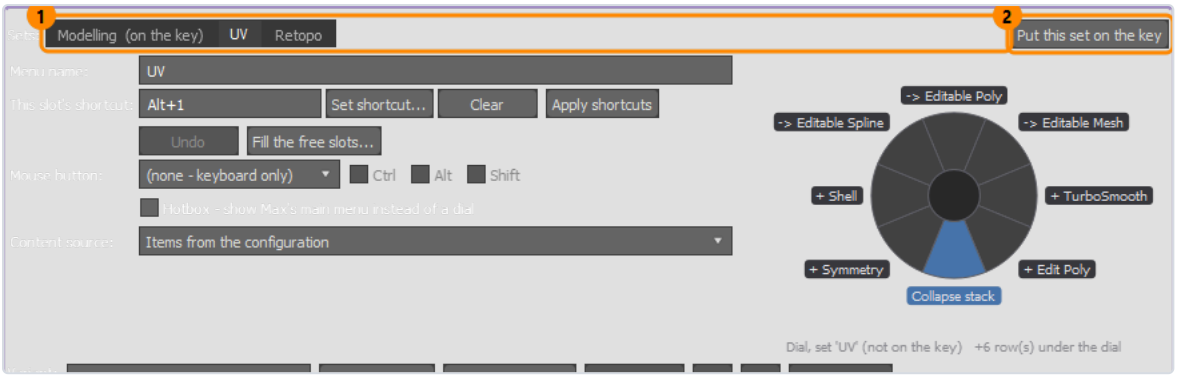
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20 Pages - several dials on one key, turned with the mouse wheel

4 min

Set (page) - another full dial kept on the same key, with its own name. Hold the key and turn the mouse wheel to go from page to page. The key, the mouse button and the other 23 dials stay as they are.

1. In the editor, click the dial - say **Menu 1** - and find **Sets of this dial** under the list of dials.
2. Press **New set...**, call it *UV* and choose **A copy of what the dial shows now** (or **An empty dial**). The new set becomes the next page. A **tab** for it appears above the dial's contents - *Sets: Modelling (on the key) | UV*.
3. Change the new set like any dial - directions, list, variants. Click a tab to see and change that set. Save.
4. In 3ds Max, hold **Alt+1** and **turn the mouse wheel**: towards you is the next page, away from you the one before, in the same place. The line under the dial says *page 2 of 2 - mouse wheel*. Let go over a command to run it. Next time you press Alt+1, the dial opens on the page you used last - even after 3ds Max restarts.



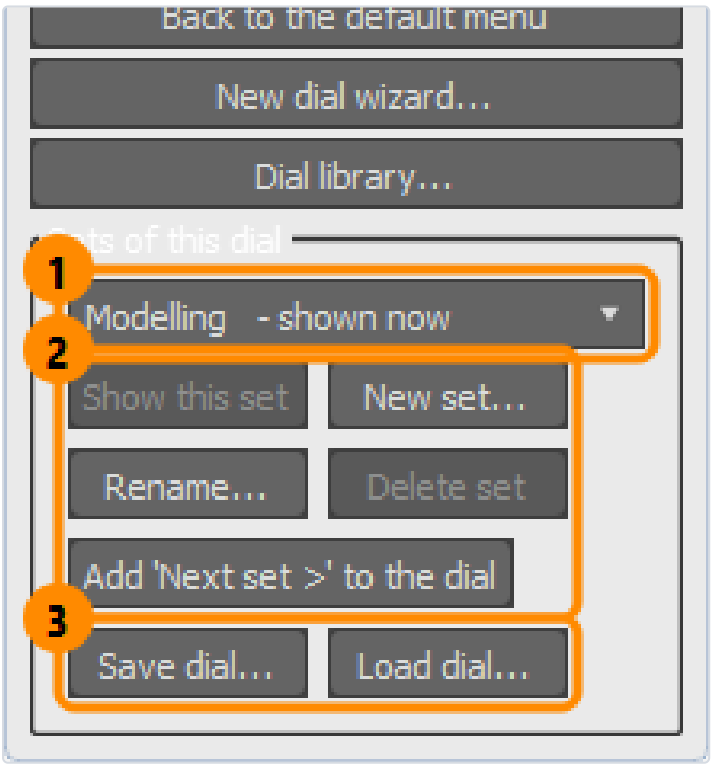
The sets of a dial as tabs (1); Put this set on the key (2).

TRY IT

Hold Alt+1, turn the wheel one step towards you, aim at a command on the UV page and let go. Press Alt+1 again: it opens on UV, where you left it. Turn back to Modelling the same way.

Which page comes first. The set marked *(on the key)* is the first page. To make another set the first one, use any of these:

1. In the editor: open the set's tab and press **Put this set on the key**. (Or pick the set in **Sets of this dial** and press **Show this set**.) Save.
2. From the dial itself: press **Add 'Next set >' to the dial**. Every set of the dial gets a *Next set* > row. Pick it, and the next set becomes the first page. Save.
3. With a key: **Customize > Hotkey Editor**, search `next set`, give **Menu 1 - next set** a key. Each of the 24 dials has one.



Sets of this dial: the sets (1), switching and managing them (2), saving and loading one dial (3).

> **When does the wheel turn pages?** Only while you hold the key of a dial that has more than one page. It also works in the hotbox, when the hotbox is one of the pages (Alt+5). In a submenu, in the search, or on a dial with one page, the wheel zooms and scrolls as usual. One step of the wheel turns one page.

> Switching from the dial or with a key is saved at once. If the editor is open with changes not saved, press **Reload from file** first.

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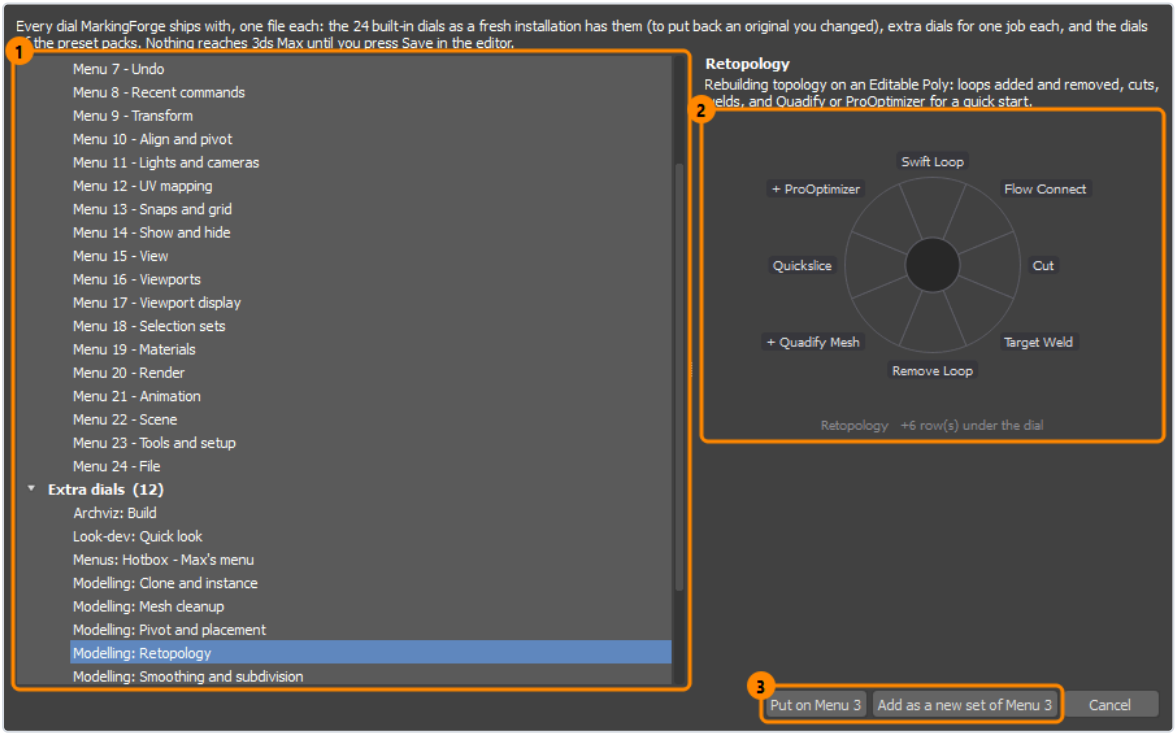
21 One dial from a file, or from the library

3 min

Any dial can be saved as a file of its own (`.mfdial`). MarkingForge comes with 51 of them in the **dial library**: the 24 dials as they come, 12 extra dials, each for one job, and the 15 dials of the preset packs.

A dial from the library

1. Click a dial in the list of dials and press **Dial library...** under it.
2. Its original is already selected. **Restore Menu N** puts it back as it came.
3. Or choose another dial. The extra ones are: Mesh cleanup, Pivot and placement, Clone and instance, Smoothing and subdivision, Retopology, Quick look, Links and helpers, Build, the viewport pages of Alt+5 and the hotbox. There are also the dials of the preset packs.
4. **Put on Menu N** swaps the dial for the one you chose. The key and mouse button stay.
5. **Add as a new set of Menu N** keeps what the dial shows, and adds the new dial as its next page.
6. Save.



The dial library: the dials in groups (1), the one chosen (2), Put on Menu N and Add as a new set of Menu N (3).

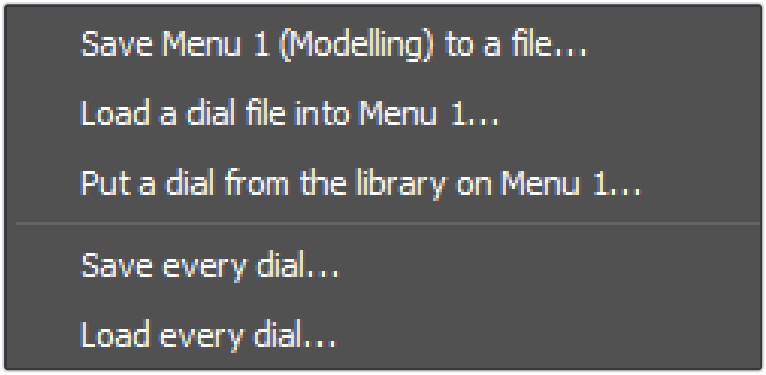
A dial from a file

1. Click the dial, press **Load dial...** in **Sets of this dial**, and choose the `.mfdial` file.
2. If the file holds scripts, **Remove the scripts** is chosen for you. Keep them only for a file you trust.
3. Choose **A new set (keeps what the dial shows now)** or **Replace what the dial shows**. Save.

Save dial... next to it saves the set chosen in the list (the open tab) to a file - to keep it, or to send it to someone.

Right-click a dial

The same is one click away: right-click the dial **in the list of dials** or **on its drawing**.



Right-click on Menu 1: save it, load a file into it, put a library dial on it - or save or load every dial.

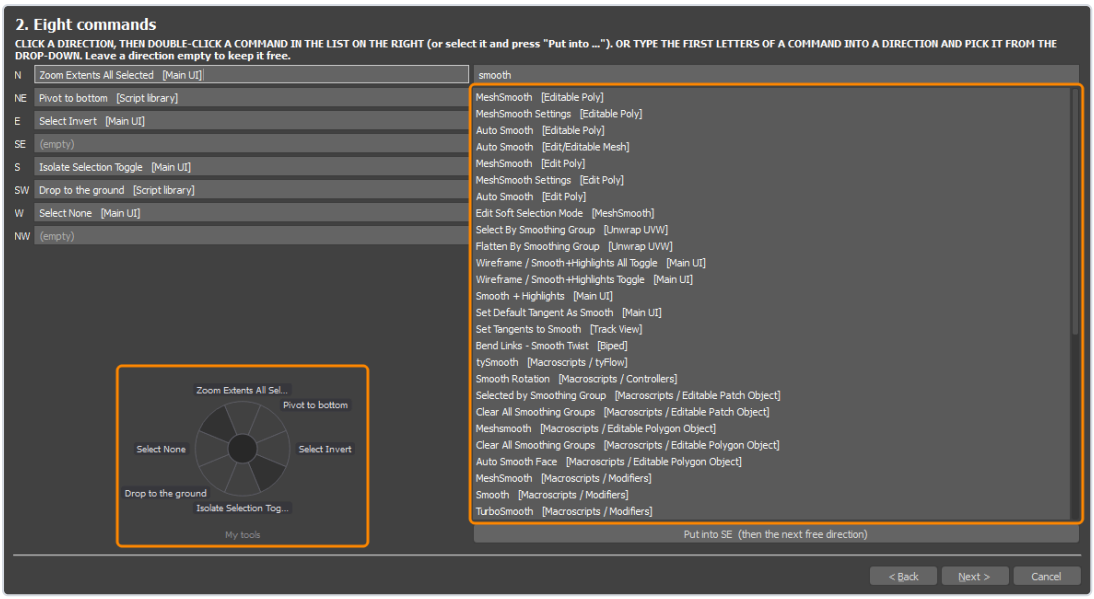
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22 A new dial in a minute - the wizard and ready-made variants

4 min

The new dial wizard

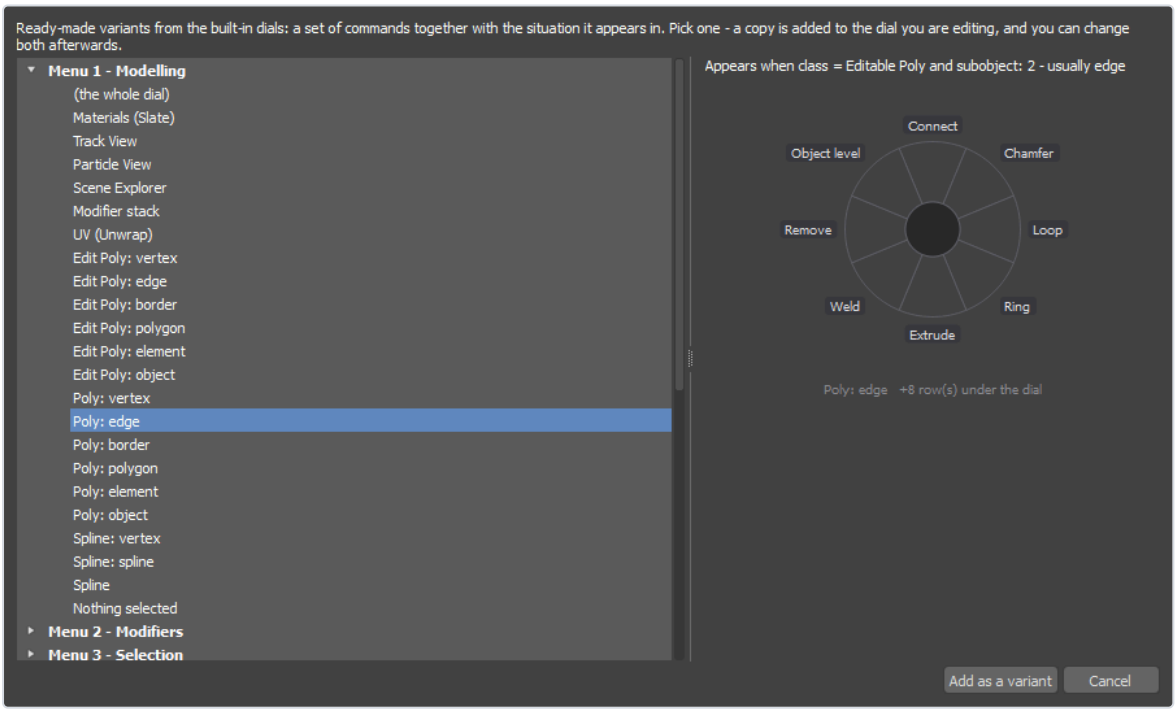
1. Press **New dial wizard...** under the list of dials.
2. Pick the dial and a name.
3. Leave **Add it as a new set of this dial - the key keeps what it opens now** switched on. Then what the dial shows now is kept, and your new dial becomes its next page.
4. For each direction, type part of a command's name and pick it from the list. 3ds Max commands and the ready-made scripts are both there. The drawing beside the fields shows the dial as you fill it. A name that matches nothing turns red.
5. Finish, then Save.



The wizard's second page: eight fields and the dial they make.

Ready-made variants

1. Click a dial and press **From a template...** next to **Add a variant...**
2. On the left: every variant the built-in dials have - Editable Poly level by level, Edit Poly, lines, cameras, lights, the windows - and each built-in dial as a whole. On the right: when it shows and what it holds.
3. Press **Add as a variant**. It comes with its situation already set (for a whole dial, you are asked for one). Change anything you like, then Save.



From a template: the variants of the built-in dials, and when each one shows.

How to learn it, and good habits

When	Do this
Day 1	Use Alt+1, Ctrl+Alt+7, Alt+3 and Alt+7, and look at the dial. Print the cheat sheet.
Days 2-5	Start flicking the five gestures of lesson 3. Add Ctrl+Alt+1 (Transform) and Ctrl+Alt+6 (Show and hide).
Week 2	Swap the directions you never use for commands you look for in menus. Add one variant for the work you do most.
Week 3	Colour the dials you use most - a ready-made look is one click. Put your other frequent commands in rows; move to a direction the ones your hand wants.
Week 4	Give your main dial a second page for a job you do now and then - UV, retopology, a dial from the library - and turn to it with the mouse wheel.

Rules for your own dials

Opposites face each other

Grow / Shrink, Hide / Unhide, Open / Save on opposite sides - one movement and its reverse.

Do not move what you know

Once your hand knows a place, leave it there. Add, do not shuffle.

One job per dial

"What do I create", "what do I hide". A dial of unrelated commands is hard to learn.

Exact variants on top

Edit Poly on vertices above Edit Poly, above Editable Poly.

Short captions

One or two words. Your hand needs the direction; your eye only a hint.

Test with a flick

If a gesture only works while you look, the command is in the wrong place.

If something goes wrong

What happens	What to do
Holding the key does nothing	**MarkingForge > Set up standard shortcuts...**, or check the key in **Customize > Hotkey Editor**.
A different dial opened than I expected	A variant for another situation showed - the object under the mouse, the level or the window decides. Move over empty space and try again.
Alt+5 shows the viewport - where is the hotbox?	Keep holding Alt+5 and turn the mouse wheel one step away from you.
A caption is red	That command is not in this 3ds Max (a plugin is missing). Put another one there in the editor.
A dial is empty	A dial that fills itself has nothing to show. Select one object for the modifiers; make named selection sets for Selection sets.
I changed something, but the dial is the same	Press **Save** in the editor.
I want all the original dials back	In the editor: **Load default dials...**
I changed one dial and want its original back	Click it in the editor, **Dial library...**, **Restore Menu N**, Save.
Turning the mouse wheel over a dial zooms the viewport	The dial has one page, or you are in a submenu or the search. The wheel turns pages only on a dial with more than one page (lesson 20).
A dial stayed on screen after I let go, and other dial keys did nothing	Sometimes 3ds Max misses that you let go (for example after starting a Text object). MarkingForge watches the key itself and closes the dial a moment later. If a dial still stays open, press Esc and send **MarkingForge > Diagnostics > Event log** with your report.
Letters I type over a dial vanish	That is **Type-to-search** (trick 5). Esc closes the search. Switch it off in Experimental features if you do not use it.
After switching panels, a dial key does nothing until I click a viewport	The keyboard got stuck on 3ds Max's menu bar (pressing Alt alone does that). MarkingForge frees it by itself. If not, press Esc or click the viewport, and send **MarkingForge > Diagnostics > Event log** with your report.
Where are the other guides?	**MarkingForge > Documentation** opens the folder with all of them.

Every option, every command and more problems with their fixes are in the **MarkingForge Reference**.

> **Still stuck?** Open an issue at github.com/Forge-Plugins/MarkingForge/issues or write to forgeplugins@gmail.com. **MarkingForge > Report a bug...** puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

Support, bug reports and feature requests

ISSUES	github.com/Forge-Plugins/MarkingForge/issues
ROADMAP	trello.com/b/7gJVqG8Y/markingforge-features-roadmap
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.